ASTANA IT UNIVERSITY

SOFTWARE ENGINEERING DIPLOMA PROJECT

TT KZ

Design and Development of the TT KZ Web Platform for Digital Integration of the Table Tennis Community in Kazakhstan

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Repository: https://github.com/TTKZ12/TTKZ_diploma_work

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1. Project proposal

1.1 Background and problem

Table tennis participation brings together players, club operators, coaches, and tournament organizers. Their activities depend on information about venues, availability, suitable partners, and competitions. Multiple disconnected channels can potentially complicate discovery and coordination. These are hypotheses to test in Kazakhstan rather than established facts about all local providers.

TT KZ proposes a unified user-facing environment for finding halls, reserving playing sessions, maintaining player profiles, discovering partners, and browsing tournament announcements. The research problem is whether a consolidated platform addresses demonstrable needs that current channels and services do not sufficiently meet in the Kazakhstani table tennis context.

1.2 Aim and research questions

Aim: To design and develop a web-based TT KZ platform intended to support the digital integration of Kazakhstan's table tennis community through centralized facility, booking, player-discovery, and tournament-information services.

ID	Research question
RQ1	What problems do Kazakhstan-based table tennis players and facility representatives experience when finding venues, booking tables, and coordinating activities?
RQ2	Which relevant functions are offered by federation, community, and booking platforms, and where do gaps remain?
RQ3	Which technologies and architecture best satisfy TT KZ functional, usability, security, and reliability requirements?
RQ4	How can TT KZ prevent overlapping confirmed reservations during simultaneous requests?
RQ5	How can usability, booking correctness, and platform value be tested against an explicit baseline?

1.3 Objectives

- Gather empirical user and operator needs through stakeholder discussions.

- Critically review literature on digital sport, table-tennis communities, facility bookings, concurrency, usability, and player matching.
- Compare existing systems using consistent criteria and verify claimed feature availability.
- Evaluate competing frontend, backend, and database choices against clear project-specific criteria.
- Define functional/non-functional requirements with testable acceptance criteria.
- Design platform context, use cases, component interactions, and persistence boundaries.
- Implement and test the five priority features over subsequent development stages.

1.4 Scope and expected outcomes

In scope (MVP target)	**Out of scope for initial pre-defense / possible later expansion**
Sports hall discovery; hall details	Online payments; equipment marketplace
Booking of table or playing session	Live-streaming or federation API integration
Player registration and profiles	Complex ratings and AI recommendations
Finding a playing partner	Built-in direct messaging
Tournament list, details, announcements	Official competition results certification / federation replacement

Expected final outcomes: a tested working web application, documented API and database schema, usable interface, automated tests, engineering documentation, and evaluation findings. At this pre-defense stage, outcomes are primarily research and design artifacts, not production implementation.

1.4.1 Operational definition of digital integration

In TT KZ, digital integration means connecting players, authorized venue staff and tournament organizers through shared records and coordinated workflows in one web platform. A venue has identifiable tables and availability intervals; a booking links a player, table and interval; an opt-in profile supports partner discovery; a tournament record links an event to its location and responsible publisher. The integration is within TT KZ and does not imply automatic synchronization with WhatsApp, 2GIS, RTTF or federation systems.

Participants	**Connected process and data**	**Observable outcome**
Players and venue staff	Venue discovery → availability → reservation → cancellation; venue_id, table_id, player_id, interval, booking status.	Less effort to identify and request a suitable slot; no overlapping confirmed records.

Participants	Connected process and data	Observable outcome
Players and other players	Opt-in profile search by city, level and availability preferences.	Users can locate a suitable visible profile without exposing private contact data.
Organizers and players	Authorized publishing and updating of tournament title, dates, location, participation conditions and source.	Users find current event information and identify its source and last update.

1.4.2 Initial pilot and inference limits

The initial pilot is limited to Astana. We propose working with one or two cooperating table tennis venues and recruiting five adult users for an exploratory usability evaluation. No venue partnership or recruitment completion is claimed. Historical interview responses are not assumed to come from Astana because participant locations require confirmation. The project title describes the wider Kazakhstan context; the present evidence does not establish nationwide demand or readiness for a national rollout.

Pilot measures are booking effort, reservation integrity and access to venue/tournament information. Section 9 defines the planned baseline and targets. Expansion beyond Astana depends on verified local data, venue participation, successful pilot evaluation and a separate assessment of transferability.

1.5 Participants and responsibilities

Participant	Primary duties
Zhanybek Alikhan	Backend architecture; PostgreSQL schema; FastAPI API; SQLAlchemy/Alembic; booking business rules and conflict prevention; validation, authentication and access control; backend tests; Docker; API documentation.
Tazhimova Araizhan	React/TypeScript frontend; UI/UX; responsive layout; frontend validation and state; API integration; client-side tests.
Both team members	User research; requirement refinement; API contract; integration/testing; GitHub reviews; YouTrack; pre-defense materials and supervisor consultation.
Zhaksylyk Tomiris	Academic supervision; review of scope, research, architecture and evaluation plan; feedback at agreed checkpoints.

2. Stakeholder evidence and provenance

2.1 Sample and evidence status

The revised dataset contains written responses from five stakeholders: three regular table tennis players (P01-P03), one coach and tournament organizer (C01), and one hall administrator (A01). P01-P03 and C01 responded on 24 October 2026; A01 responded on 25 October 2026, after the initial deadline. Four real participant exchanges were therefore available by 24 October. The earlier illustrative A01 answers were not empirical data and are replaced in this revision. The sample is exploratory, not representative.

Data were collected through written, question-guided messaging exchanges: Telegram for P01-P03 and C01, and WhatsApp for A01. Tazhimova Araizhan conducted the exchanges; original responses were in Russian and Zhanybek Alikhan translated them into English. Metadata and agreement to anonymized use are reported by the team; original screenshots remain to be inserted in Appendix E. No audio recording was made. Participant locations and current translation checks remain to be documented.

Code and role	Reported perspective	Provenance status
P01 – player	Plays about once a week; describes uncertainty about slots and delayed replies.	24 Oct 2026; written Telegram exchange. Current translation/source check pending; agreement to anonymized use reported.
P02 – player	Plays daily; reports good venue access and uses RTTF/TTRanking.	24 Oct 2026; written Telegram exchange. Current translation/source check pending; agreement to anonymized use reported.
P03 – player	Plays around six times a week; mentions price and suitable time.	24 Oct 2026; written Telegram exchange. Current translation/source check pending; agreement to anonymized use reported.

Code and role	Reported perspective	Provenance status
A01 – hall administrator	Telephone/WhatsApp requests; administrator checks availability and confirms a table; alternative slots offered when full.	25 Oct 2026; written WhatsApp exchange. Current translation/source check pending; agreement to anonymized use reported.
C01 – coach or organizer	Fragmented announcements, repeated questions and disconnected tools.	24 Oct 2026; written Telegram exchange. Current translation/source check pending; agreement to anonymized use reported.

2.2 Thematic findings and negative cases

Venue discovery and reservations. P01 and P03 use 2GIS; P01 also relies on friends and chats. Both describe telephone or message contact for reservations. P01 reports uncertainty about availability and response delays; P03 mentions financial limitations and a suitable time. In contrast, P02 reports no difficulties and good facility availability. The proposed booking workflow addresses an observed convenience issue in part of this small sample, not universal dissatisfaction.

Partner and tournament discovery. All three players use tournaments, friends or community chats to find comparable opponents. Tournament news comes from chats, acquaintances, social media and RTTF. P01 reports not having encountered a single amateur calendar; this does not establish that no such calendar exists.

Venue operations. A01 reports telephone and WhatsApp bookings, usually in advance but also on the day when tables are available. The administrator controls availability for approximately eight rental tables, checks the date, time and number of guests, and assigns a table after confirmation. When demand exceeds available tables, another time or date is offered. This supports explicit availability checks and confirmation states; it does not establish actual double bookings, spreadsheet use or delays caused by multiple staff. Customers ask about availability, specific tables, payment and cancellation or rescheduling. Payment is made on arrival without a booking deposit. No separate substantive answer identified the most time-consuming tasks or online-booking limitations.

Tournament communication. C01 reports posting announcements through WhatsApp, Instagram and Telegram and using Google Forms and spreadsheets for other tasks. Missed updates and repeated questions motivate a maintained event listing. Registration and notifications remain possible later enhancements, not current implementation claims.

Information quality. Participants mention table condition, lighting, space, prices, schedule and amenities. P03 names Unreal Engine 5 as an organization tool; this response is retained but

excluded from conclusions about table tennis booking applications because it appears to reflect a different interpretation of the question.

2.3 Evidence to requirements

Evidence	Requirement or design implication	Limitation
P01, P03: map search and direct venue contact	FR-02 and FR-03: venue information and available-slot booking.	No verified time-saving measure yet.
P01: unclear slots; A01: availability checked before confirming competing requests.	FR-04 and NFR-02: atomic non-overlap rule and conflict recovery.	Confirmed double bookings and their frequency were not established. Atomic integrity is an engineering requirement.
P01–P03: social ways of finding opponents	FR-05 and FR-06: private-by-default, opt-in player profiles.	No proof that users will adopt a new platform.
P01–P03 and C01: distributed tournament information	FR-07 and FR-11: maintained listings and authorized updates.	Coverage depends on source and organizer cooperation.
A01: administrator controls availability; customers ask about cancellation and rescheduling.	FR-08: manager maintenance. FR-09: clarify and display cancellation policy.	Specific cancellation rules and management-interface preferences were not confirmed.

2.4 Interpretation

The responses provide preliminary motivation for evaluating integrated discovery, booking and information workflows. They do not establish nationwide demand, willingness to pay, venue adoption or platform effectiveness. Statements above remain stakeholder reports with pending provenance checks. If a source cannot be authenticated, its quotation and requirement support must be downgraded or removed; a new session must be dated as a new interview rather than backdated. Only one administrator was consulted, after the initial deadline. Repeated messages were consolidated and were not counted as additional participants or findings. A01 did not provide separate substantive answers about time-consuming tasks or online-booking limitations.

3. Critical literature review

3.1 Review scope and selection

This purposive review synthesizes 13 scientific publications from 2020–2026. The narrative, synthesis matrix and scientific bibliography contain the same 13 studies. Most studies are recent; Gong et al. (2020) is retained as a foundational matching-method reference. Official technical and product sources are listed separately and are not counted as scientific publications. This is a critical project review, not a systematic review conducted by the team.

3.2 Digital transformation and national context

Dašić (2026) provides a conceptual sports-management framing of digital transformation as an organizational and stakeholder process, rather than adoption of isolated IT tools. Wan et al. (2025), using sector-level Chinese economic time series, report varying links between digital industry development and sports service subsectors. These papers support examining integrated sport services, but neither demonstrates demand for a particular Kazakhstan table tennis booking platform.

Ibrayeva et al. (2025) surveyed adults across Kazakhstan and reported low physical activity for 19.7% of analyzed respondents and no leisure-time activity for 58.1%. This supplies Kazakhstan-specific participation context but is not evidence about table-tennis demand, user willingness to book online, or app effectiveness. The cross-sectional design and self-report measures constrain causal inference.

3.3 Clubs and community interaction

An et al. (2025) investigated amateur table-tennis club management in Guangzhou using questionnaires and stakeholder interviews. Their CPECDS framework identifies consumer needs, competition organization, digital technology, and other operational factors, supporting TT KZ's decision to involve clubs as well as players in requirements discovery. Its organizational findings cannot be transferred uncritically to Kazakhstan.

Stegmann et al. (2026) experimentally studied platform features and social capital in sports-fan engagement. Exploratory associations for different engagement modes illustrate possible roles for broader connections and group affiliation; overall multivariate findings were not statistically decisive. This is conceptual justification for player and community functionality rather than proof of behavioral change among recreational table-tennis players.

3.4 Reservation workflows and data consistency

German et al. (2024) propose Appoint Deportes, an online sports-facility reservation system grounded in observations, interviews, and process/interface modeling. The research illustrates facility discovery, schedule visibility, and booking flows but offers limited evidence of operational reliability under concurrency. Hermawan and Suharso (2025) test atomic Firebase transactions in a billiard-table reservation case. In ten small concurrent-user scenarios, the authors report double

bookings before the control and none after, although the trials cover only 2–11 users and omit wider production performance evaluation.

For TT KZ, it is crucial to distinguish front-end availability presentation from a database-enforced non-overlap invariant. Since the planned database is PostgreSQL, an initial design option is range-aware exclusion constraints combined with transactional API operations and appropriate conflict responses. This approach is supported by PostgreSQL technical documentation, not directly tested by Hermawan and Suharso's Firebase experiment.

3.5 User-centered interface design

Maringka and Lumingkewas (2024) use Design Thinking and Maze usability testing for a sports-facility rental prototype. Among 51 testers, the reported direct success rate was 78.4%, average task duration 46.1 seconds, and misclick rate 30.2%. Differences between screen scores indicate that user-centered design requires repeated measurement and adjustment rather than relying on visual attractiveness alone.

3.6 Player matching

Gong et al. (2020) propose OptMatch, a computational model of competitive matchmaking informed by historical match interactions and predictive optimization. It is scientifically strong but concerns esports and competitive team formation rather than local offline partner search. TT KZ should initially explore simple opt-in profile filters based on location, level and availability; complex data-driven ranking would require sufficient consented match histories and separate evaluation.

Additional perspectives on service integration and measurement

Anantha Sekoh et al. (2025) describe development of a mobile sports-field booking application using agile development and interviews with 20 managers and users. This supports iterative requirements discovery and scheduling workflows. Reported benefits in that setting are not treated as causal evidence that TT KZ will increase participation or reduce booking effort in Astana.

Saputra and Asmawati (2025) review integrated management information systems in sports facilities, drawing attention to information access, staff adoption, security and privacy. Their small review supports connecting user-facing information and venue operations, but does not validate a nationwide community platform.

Rai et al. (2025) synthesize social-media research in sport using the TCCM framework and identify geographical and theoretical gaps. This helps explain why evidence about fan engagement or communication channels cannot simply be transferred to local offline players. TT KZ will assess specific task outcomes rather than assume community growth.

Iversen et al. (2026) distinguish bookings, actual usage and attendance in sports-facility measurement. Accordingly, TT KZ will not label booked capacity as attendance or claim increased physical activity from reservation records alone. Actual attendance is outside the first pilot unless a venue provides a consented verification process.

3.7 Literature synthesis matrix

Study	Method / evidence	Key transferable insight	Limit / TT KZ interpretation
Dašić (2026)	Conceptual literature review	Integrated sport services and stakeholder networks	Not empirical validation of TT KZ
Stegmann et al. (2026)	Scenario experiment, 226 sports fans	Design can shape types of engagement	Effects exploratory; fans not players
Wan et al. (2025)	VAR/economic time series, China	Digital sport services vary by subsector	Macro outcomes not user needs
German et al. (2024)	Observations, interviews, system design	Facility discovery and reservation flows	No rigorous conflict-prevention test
Hermawan & Suharso (2025)	Ten before/after concurrency scenarios	Atomic booking operations needed	Firebase; only 2–11 concurrent users
Maringka & Lumingkewas (2024)	Design Thinking; 51 prototype testers	Task-oriented usability metrics	Prototype and remaining usability issues
An et al. (2025)	377 club consumers and interviews	Clubs' needs, events, digital operations	Guangzhou management focus
Ibrayeva et al. (2025)	Nationwide cross-sectional survey	Kazakhstan participation context	Not table-tennis or digital-service study
Gong et al. (2020)	Models and four datasets	Skill and interaction-aware matching	Advanced esports/team setting
Anantha Sekoh et al. (2025)	Agile development; interviews with 20 facility stakeholders	Iterative booking workflow design	Different sport/facility setting; no local effect established.
Saputra & Asmawati (2025)	Review of seven IMIS studies	Connect information, operations and privacy	Small review; facility scope rather than complete sports community.
Rai et al. (2025)	Systematic synthesis of 203 papers	Evaluate channels and local context	Social-media literature does not establish local platform adoption.
Iversen et al. (2026)	Conceptual and empirical facility measurement	Distinguish booking from usage/attendance	Different facility context; no TT KZ evaluation.

3.8 Research gap and intended contribution

Across the reviewed literature, reservation correctness, sports-community engagement, interface usability, and table-tennis club management are each studied, but the selected studies do not validate a single Kazakhstan-focused integration of hall discovery, bookings, partner search, and tournament information. That observation about THIS REVIEW is not evidence that no such market product exists. The proposed contribution is a locally validated combination of those workflows with explicit booking integrity and measurable usability, to be tested rather than presumed successful.

Count reconciliation: 13 studies are discussed in the narrative, 13 appear in the synthesis matrix, and 13 appear in the scientific reference list. Dašić and Wan are included in all three. External validity remains limited: none of these papers establishes demand for the Astana pilot.

4. Existing system comparison

4.1 Scope and evidence rules

We compare three external benchmarks: ITTF, TTFRK and RTTF against nine consistent criteria. TT KZ is the proposed solution, not a fourth existing system. Public-page observations are distinguished from authenticated feature tests. “Not established” means that the reviewed pages do not provide sufficient evidence; it does not assert that a capability is absent. Open TT is excluded from this revision to keep the comparison focused on benchmarks the team can readily demonstrate.

Criterion	ITTF	TTFRK	RTTF	TT KZ target
Event information	Published event news	Calendar verified	Tournament listings	Must
Ratings or rankings	World rankings verified	Downloadable lists verified	Rating functions visible	Out of MVP
Player information	Ranking lists	National-team information links	Public player links	Opt-in profiles
Venue discovery	Not established	Not established	Club directory visible	Astana venue catalog
Table booking	Not established	Not established	Not established	Must
Partner discovery	Not established	Not established	Not established	Opt-in city/level search
Tournament announcements	Event news	Dated news verified	Announcements visible	Must

Criterion	ITTF	TTFRK	RTTF	TT KZ target
Venue schedule editing	Not established	Not established	Not established	Authorized manager
Geographic orientation	International	Kazakhstan federation	Russia and regional communities	Astana pilot

4.2 Dated comparison evidence

Evidence ID	Official public source	Observation on 29 October 2026
SYS-01	https://ttfrk.kz/ru	Federation identity, dated news, sporting events and national-team links.
SYS-02	https://ttfrk.kz/ru/calendar	2026 event entries list dates and locations, including Astana events.
SYS-03	https://ttfrk.kz/ru/rating	2026 monthly and age-category ranking-download links. File correctness was not independently tested.
SYS-04	https://ttfrk.kz/ru/teams	Links to national-team information; not proof of editable community profiles.
SYS-05	https://www.ittf.com/rankings/	Official indexed ranking page and weekly lists; direct retrieval was restricted in this check.
SYS-06	https://rttf.ru/	Official indexed homepage shows club directory, player links, tournament and rating information; direct retrieval timed out.

TTFRK is therefore verified for public calendar, announcement and ranking publication functions. Public-page inspection does not establish a self-service booking, partner-filter or manager interface. The product gap is a testable local-workflow hypothesis, not a claim that these services do not exist elsewhere. Dated source links and observations are preserved in the companion evidence register; no authenticated usability or booking test of these sites is claimed.

5. Technology alternatives and justification

The team selected React with TypeScript for the frontend, FastAPI for the backend and PostgreSQL for data storage. The comparisons below explain these decisions against the project scope and existing work. The frontend prototype exists; backend and database implementation and validation remain pending. No benchmarking results or supervisor approval of the stack are claimed.

Decision	Options	Working rationale	Decision and validation status
Frontend framework	React + TypeScript; Vue + TypeScript; Angular	React is appropriate for reusable interactive components and team familiarity; Vue is approachable; Angular is more opinionated and larger in scope.	React + TypeScript selected. Frontend prototype implemented; backend integration pending.
Backend framework	FastAPI; Django + DRF; NestJS	FastAPI gives Python typing, validation and OpenAPI tooling. Django offers strong built-in management; NestJS aligns with TypeScript across layers.	FastAPI selected. Runnable backend implementation pending.
Database	PostgreSQL; MySQL; MongoDB	Bookings and users form relational data. PostgreSQL supports ACID transactions, foreign keys and exclusion constraints for overlapping time ranges.	PostgreSQL selected. Schema implementation and concurrency tests pending.

5.1 Persistence, tooling and engineering boundaries

SQLAlchemy is proposed as the Python ORM, Alembic for schema migrations, Docker for repeatable local environments, GitHub for change tracking, and YouTrack for task traceability. The first iteration should prefer a modular single-backend application and one authoritative PostgreSQL database. Microservices, event streaming and advanced AI matchmaking are not initial requirements.

Potential booking invariant: for each physical table, no two active/confirmed bookings may overlap in time. One PostgreSQL option is a GiST exclusion constraint over table identifier equality and a time-range overlap operator, with care over time zones, status transitions, cancellations and migration compatibility. API conflict responses and concurrent test cases must complement database enforcement (PostgreSQL Global Development Group, n.d.).

Decision criteria are functional fit, booking integrity, implementation effort, maintainability and reuse of existing work. The following table records the selected options and trade-offs. The

general committee feedback forwarded by the supervisor permits continuation of the topic; detailed review of these technology decisions remains pending.

Decision	Comparative assessment	Selected option
Frontend	React preserves the existing component direction; Vue is a reasonable lightweight alternative but requires adaptation; Angular provides a more prescribed structure with additional setup.	React + TypeScript; reuse the existing layout.
Backend	FastAPI fits the Python backend role and API-first workflow; Django + DRF offers integrated administration; NestJS fits a TypeScript backend but changes the assigned language.	FastAPI; prioritize validation and a documented REST contract.
Database	PostgreSQL supports relational integrity and range exclusion constraints; MySQL supports relational transactions but requires a different overlap-control design; MongoDB requires additional modeling for relational booking rules.	PostgreSQL; validate overlap constraints through integration tests.

6. Initial requirements specification

6.1 Functional requirements

ID	Priority	Requirement	Acceptance criterion (draft)
FR-01	Must	User registration and login	Unique valid email and password create an account; duplicate email is rejected. Valid credentials authenticate; invalid credentials and unauthenticated protected requests are denied.
FR-02	Must	Sports hall listing and detail pages	Filter halls by city; selecting a hall displays its location, facilities and available schedule. No matches produce an explicit empty state.
FR-03	Must	Table booking	For an authenticated user, a future interval with start before end and inside availability creates one persisted booking ID. Invalid intervals are rejected.
FR-04	Must	Booking conflict prevention	Concurrent requests for the same table and overlapping intervals cannot both confirm. The rejected request receives HTTP 409; adjacent non-overlapping intervals remain valid.

ID	Priority	Requirement	Acceptance criterion (draft)
FR-05	Must	Player profiles	Owner can edit city, level, preferences and visibility. Another user cannot modify the profile; private contact fields are not returned publicly.
FR-06	Must	Partner discovery	Filter opted-in player profiles by city and skill level; opted-out profiles never appear. Empty results are explained.
FR-07	Must	Tournament information	Tournament list opens details containing title, date, location and organizer information; missing optional data is labeled.
FR-08	Must	Facility administrator maintenance	Assigned hall manager can update their own venue, tables and future availability only. Existing confirmed bookings are preserved; unauthorized changes are denied.
FR-09	Should	Reservation cancellation	Proposed policy: owner can cancel a confirmed booking before its start. Status becomes cancelled and the interval is released; another user cannot cancel it.

ID	Priority	Requirement	Acceptance criterion (draft)
FR-10	Should	Search/filter feedback	Search and booking views expose loading, empty, success and recoverable error states; retry does not create duplicate bookings.
FR-11	Must	Tournament publishing and source maintenance	Authorized organizer or platform administrator creates or updates event title, dates, location and conditions. Store source and last-update time; reject edits by unrelated users.

6.2 Non-functional requirements

ID	Area	Working acceptance metric / method
NFR-01	Security (Must)	All protected endpoints enforce authentication and appropriate object-level authorization; test denied cases.
NFR-02	Integrity (Must)	Zero overlapping active bookings after a documented burst test in a controlled environment.
NFR-03	Usability (Should)	At least 80% of pilot participants complete hall-to-booking prototype task without assistance (proposed threshold).
NFR-04	Performance (Should)	Proposed target: p95 <= 1000 ms and unexpected error rate < 1% for read endpoints at 20 concurrent users over 5 minutes; log p50 and p95. See section 9 for conditions.

ID	Area	Working acceptance metric / method
NFR-05	Maintainability (Must)	A clean checkout starts using documented commands and example configuration; migrations initialize an empty database; all mandatory acceptance tests pass.
NFR-06	Privacy (Must)	Collect minimal profile data; provide opt-in visibility and avoid exposing private contact fields.
NFR-07	Information freshness (Must)	Public venue and tournament details display source/owner and last-update date; stale or unconfirmed schedule data is labeled and cannot be presented as bookable availability.

The criteria above are proposed acceptance targets, not measured results. Validate priorities and the cancellation policy with stakeholders and the supervisor. Map FR/NFR IDs to screens, API operations, YouTrack tasks and tests.

7. Architecture and diagrams

High-level proposed flow: Browser (React/TypeScript) → HTTPS REST API (FastAPI) → service/business logic → SQLAlchemy → PostgreSQL. Alembic controls schema evolution; Docker Compose may support local development. This is a proposed logical structure rather than verified implemented code.

Required artifact	Contents to include	Status
System Context Diagram	Players, guests, venue managers and tournament publishers; TT KZ boundary and shared pilot workflows.	Included in Appendix C.1. Editable source and readable export are available in docs/diagrams/.
Use Case Diagram	Registration, venue search, booking/cancel, profile editing, partner search, tournament viewing and management.	Included in Appendix C.2. Editable source and readable export are available in docs/diagrams/.

Required artifact	Contents to include	Status
Component Diagram (draft)	React UI, FastAPI routes, auth, booking, venue, player, tournaments, database; interface contracts.	Included in Appendix C.3. Editable source and readable export are available in docs/diagrams/. The diagram describes the proposed architecture.

Suggested domain model for later ERD: User, PlayerProfile, Venue, Table, AvailabilityRule, Booking, Tournament and optionally TournamentOrganizer. Relationships, constraints and permissions must be reviewed before implementation; this report does not claim a completed schema.

Editable draw.io sources and readable PNG exports accompany this revision. Appendix C retains readable diagrams; optional replacement slots for draw.io screenshots and repository links are provided in Appendix H. Repository publication remains pending.

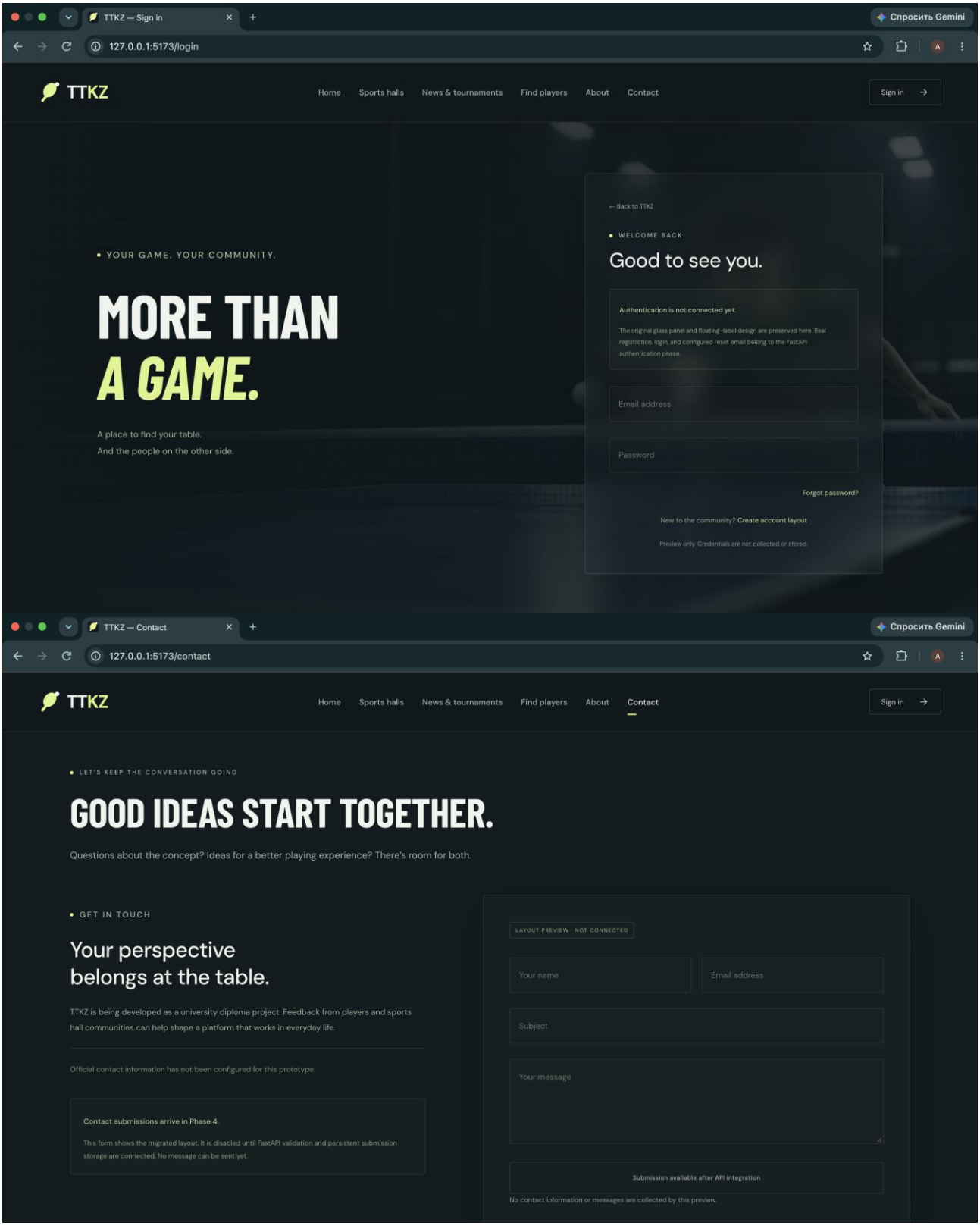
7.1 Integration traceability

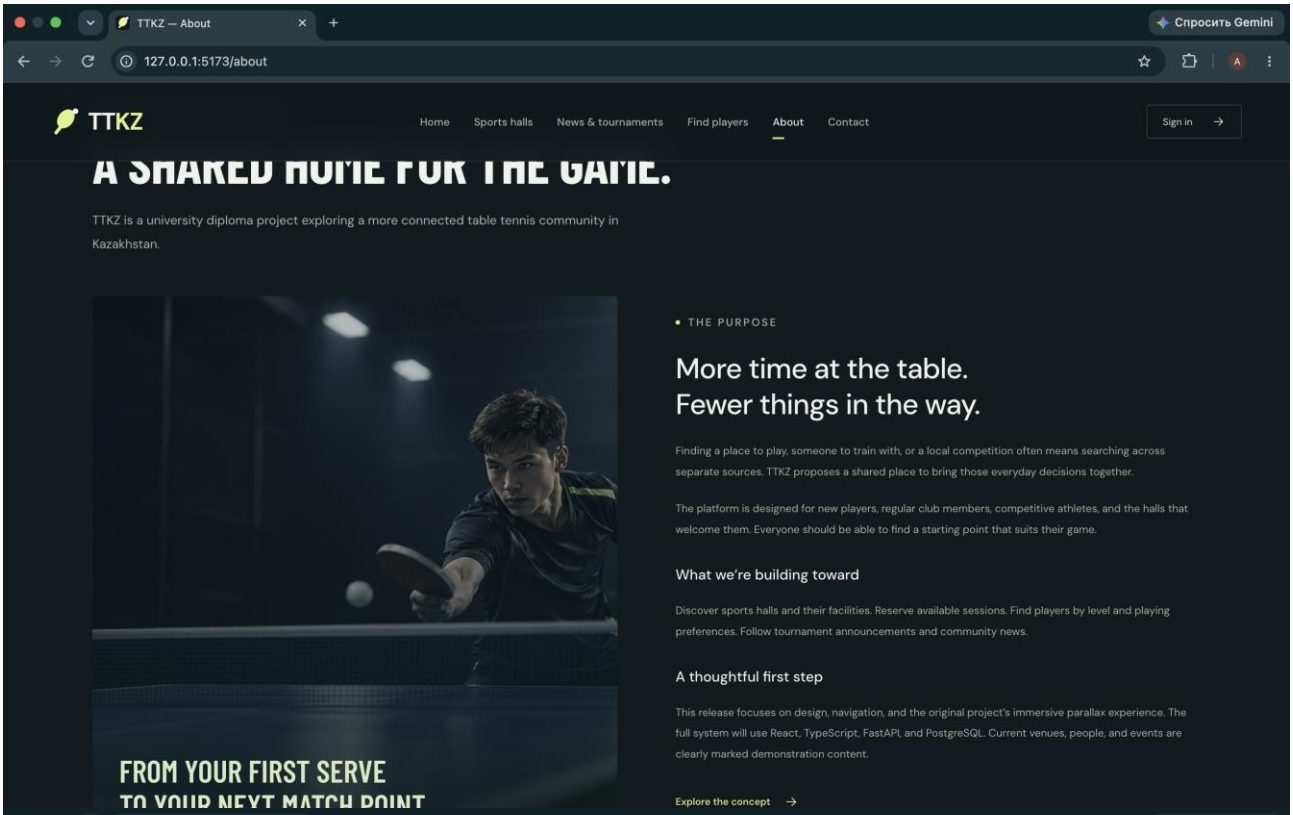
Workflow	Requirements	Components and shared records	Evaluation
Venue to confirmed booking	FR-02,03,04,08,09; NFR-02	Venue and booking services; Venue → Table → Booking → User.	Time/steps to book; 0 overlaps; 409 recovery.
Opt-in partner discovery	FR-05,06; NFR-06	Profile/search service; User and PlayerProfile visibility.	Find suitable profile; private fields never returned.
Tournament information updates	FR-07,11; NFR-07	Tournament service; Tournament, source, venue/location, publisher and update time.	Locate event details; authorized edits visible; freshness recorded.

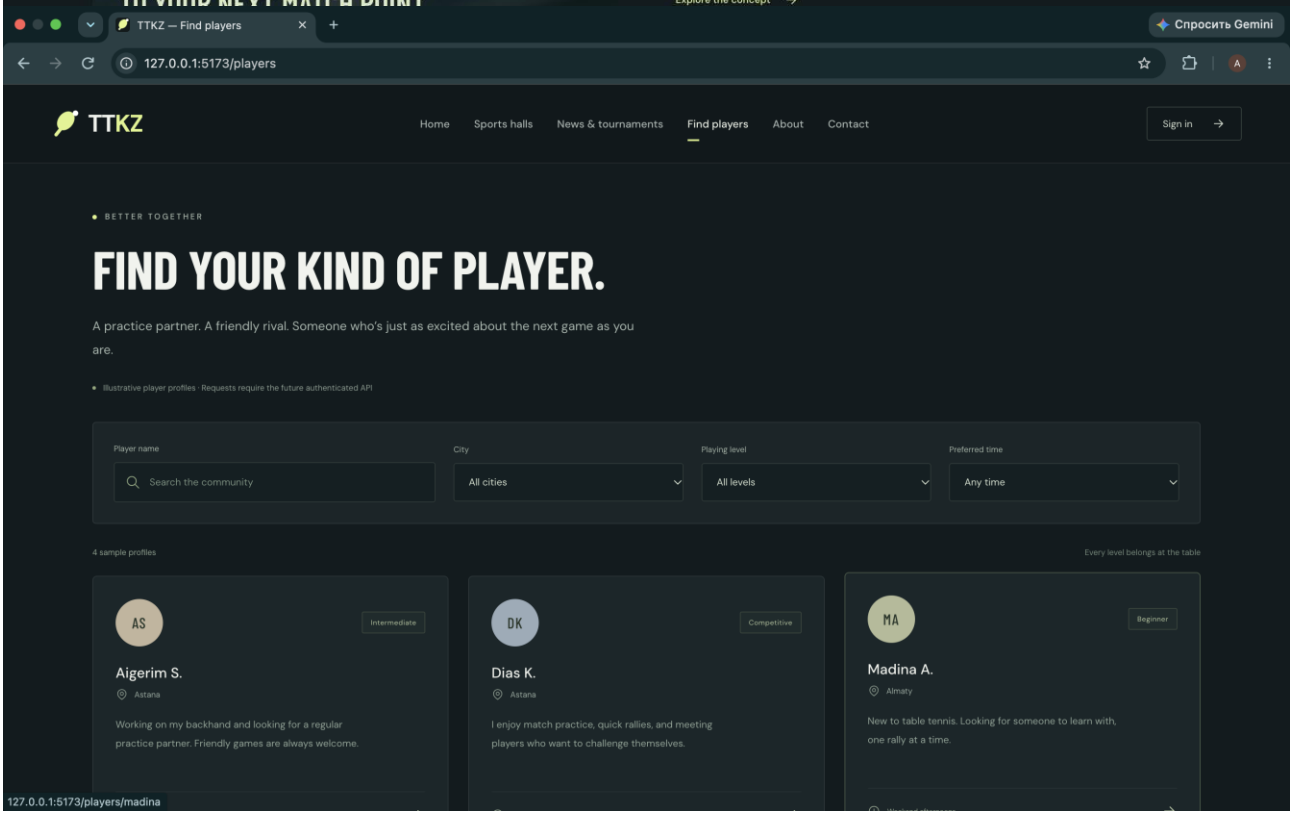
8. UX/prototype evidence

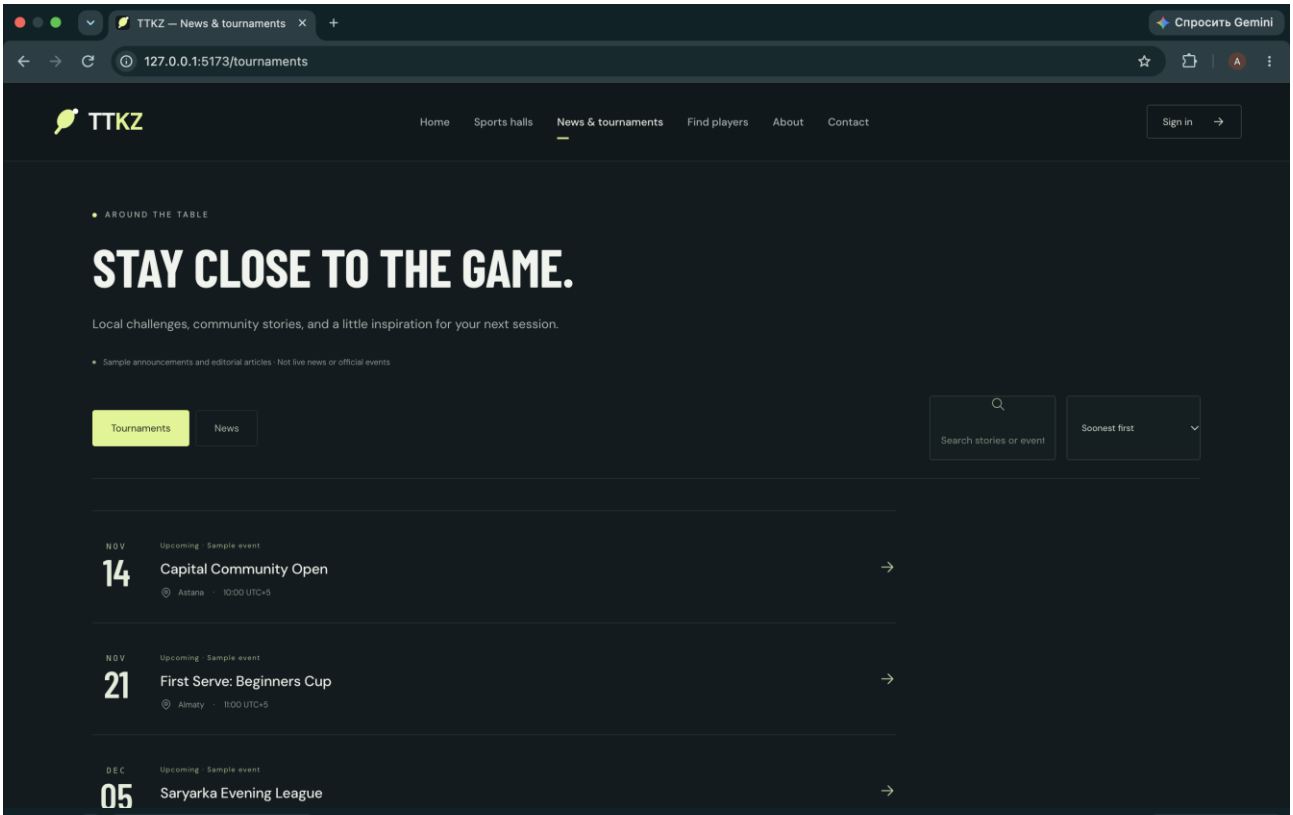
The existing screenshots document frontend layouts using demonstration data. No database or API is connected; displayed availability and confirmation states do not establish real reservations. The team reports screenshots for venue, player and tournament views, time selection and confirmation, but no frontend conflict-state screenshot. Section 8.1 includes a separate illustrative conflict mockup; it is not an implemented screen.

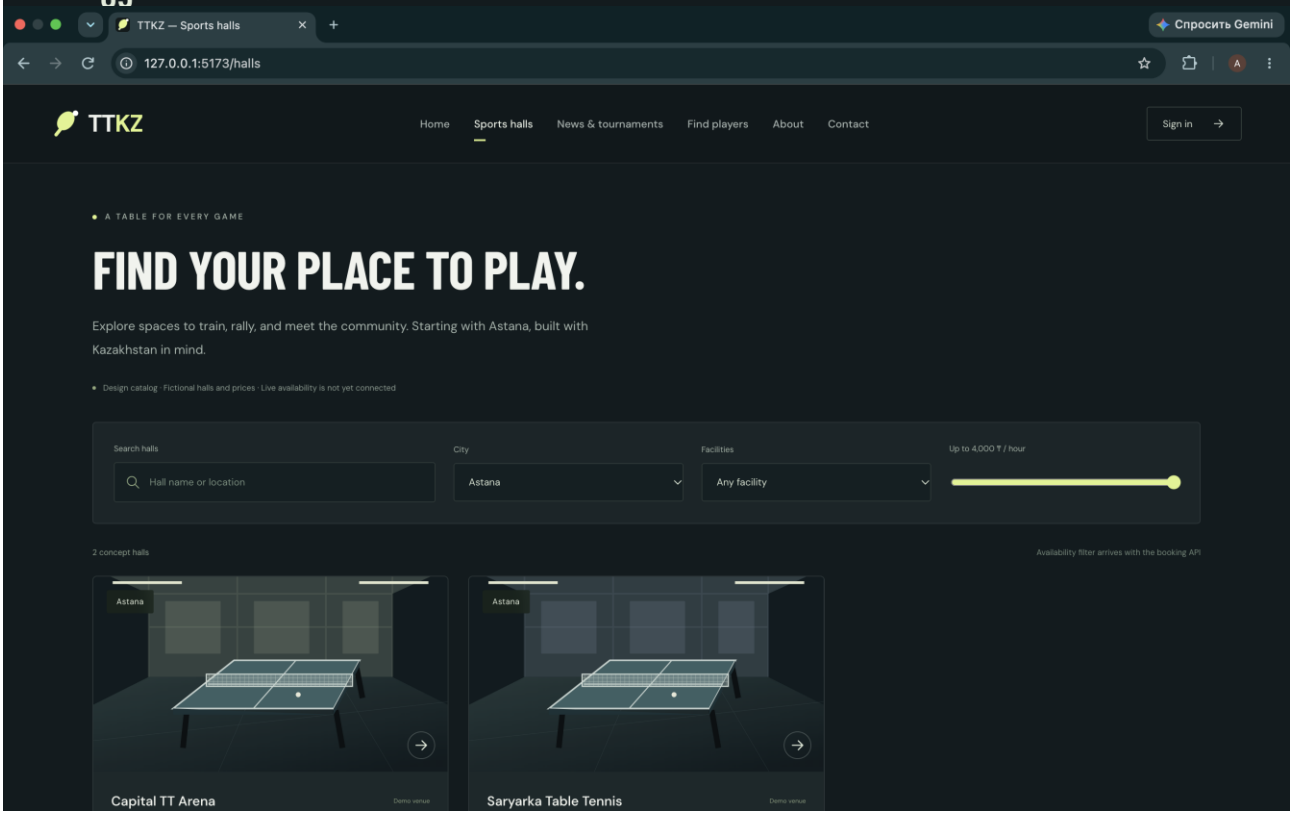
Scenario	Screens / evidence	Status
Venue discovery	Home, searchable catalog, map/list, venue details	Venue discovery layout included below; venue detail and availability flow require completion.
Reserve table	Date/time selection, conflict and confirmation states	Time-selection and confirmation screenshots reported by team; conflict is an illustrative mockup only. DB/API integration absent.
Player interaction	Profile, discover players, level/availability filtering	Player discovery layout included; own-profile editing and functional filters require verification.
Tournament discovery	List and tournament detail page	Tournament list layout included; detail flow requires verification.

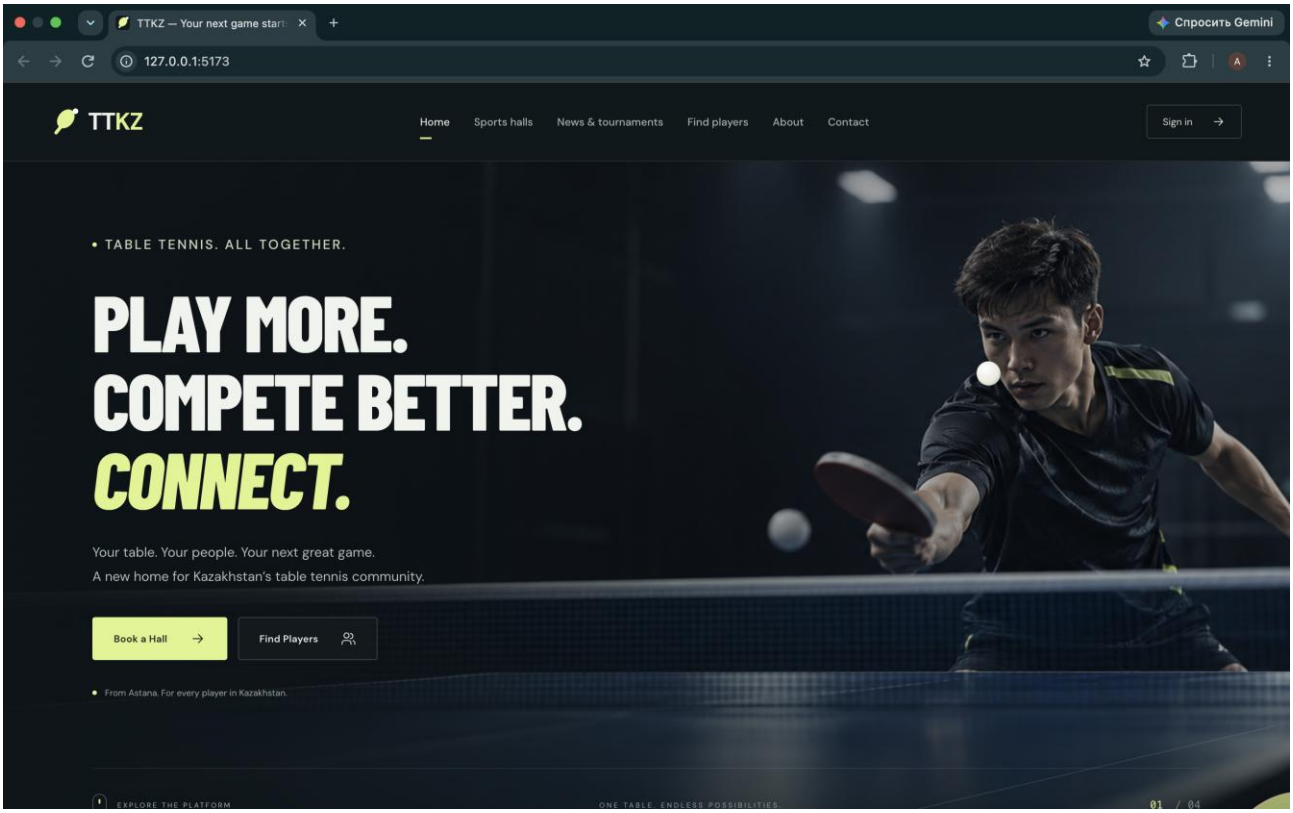

TTKZ — Sign in
Спросить Gemini
127.0.0.1:5173/login
TTKZ
Home
Sports halls
News & tournaments
Find players
About
Contact
Sign in
YOUR GAME. YOUR COMMUNITY.
MORE THAN
A GAME.
A place to find your table.
And the people on the other side.
← Back to TTKZ
WELCOME BACK
Good to see you.
Authentication is not connected yet.
The original glass panel and floating-label design are preserved here. Real registration, login, and configured reset email belong to the FastAPI authentication phase.
Email address
Password
Forgot password?
New to the community? Create account layout
Preview only. Credentials are not collected or stored.
TTKZ — Contact
Спросить Gemini
127.0.0.1:5173/contact
TTKZ
Home
Sports halls
News & tournaments
Find players
About
Contact
Sign in
LET'S KEEP THE CONVERSATION GOING
GOOD IDEAS START TOGETHER.
Questions about the concept? Ideas for a better playing experience? There's room for both.
GET IN TOUCH
Your perspective
belongs at the table.
TTKZ is being developed as a university diploma project. Feedback from players and sports hall communities can help shape a platform that works in everyday life.
Official contact information has not been configured for this prototype.
Contact submissions arrive in Phase 4.
This form shows the migrated layout. It is disabled until FastAPI validation and persistent submission storage are connected. No message can be sent yet.
LAYOUT PREVIEW · NOT CONNECTED
Your name
Email address
Subject
Your message
Submission available after API integration
No contact information or messages are collected by this preview.


TTKZ — About
Спросить Gemini
127.0.0.1:5173/about
TTKZ
Home
Sports halls
News & tournaments
Find players
About
Contact
Sign in
A SHARED HOME FOR THE GAME.
TTKZ is a university diploma project exploring a more connected table tennis community in Kazakhstan.
FROM YOUR FIRST SERVE
THE PURPOSE
More time at the table.
Fewer things in the way.
Finding a place to play, someone to train with, or a local competition often means searching across separate sources. TTKZ proposes a shared place to bring those everyday decisions together.
The platform is designed for new players, regular club members, competitive athletes, and the halls that welcome them. Everyone should be able to find a starting point that suits their game.
What we're building toward
Discover sports halls and their facilities. Reserve available sessions. Find players by level and playing preferences. Follow tournament announcements and community news.
A thoughtful first step
This release focuses on design, navigation, and the original project's immersive parallax experience. The full system will use React, TypeScript, FastAPI, and PostgreSQL. Current venues, people, and events are clearly marked demonstration content.
Explore the concept


TTKZ — Find players
Спросить Gemini
127.0.0.1:5173/players
TTKZ
Home
Sports halls
News & tournaments
Find players
About
Contact
Sign in
BETTER TOGETHER
FIND YOUR KIND OF PLAYER.
A practice partner. A friendly rival. Someone who's just as excited about the next game as you are.
Illustrative player profiles · Requests require the future authenticated API
Player name
Search the community
City
All cities
Playing level
All levels
Preferred time
Any time
4 sample profiles
Every level belongs at the table
AS
Intermediate
Aigerim S.
Astana
Working on my backhand and looking for a regular practice partner. Friendly games are always welcome.
DK
Competitive
Dias K.
Astana
I enjoy match practice, quick rallies, and meeting players who want to challenge themselves.
MA
Beginner
Madina A.
Almaty
New to table tennis. Looking for someone to learn with, one rally at a time.
127.0.0.1:5173/players/madina


TTKZ — News & tournaments
Спросить Gemini
127.0.0.1:5173/tournaments
TTKZ
Home
Sports halls
News & tournaments
Find players
About
Contact
Sign in
AROUND THE TABLE
STAY CLOSE TO THE GAME.
Local challenges, community stories, and a little inspiration for your next session.
Sample announcements and editorial articles · Not live news or official events
Tournaments
News
Search stories or event
Soonest first
NOV
14
Upcoming · Sample event
Capital Community Open
Astana · 10:00 UTC+5
NOV
21
Upcoming · Sample event
First Serve: Beginners Cup
Almaty · 11:00 UTC+5
DEC
05
Upcoming · Sample event
Saryarka Evening League

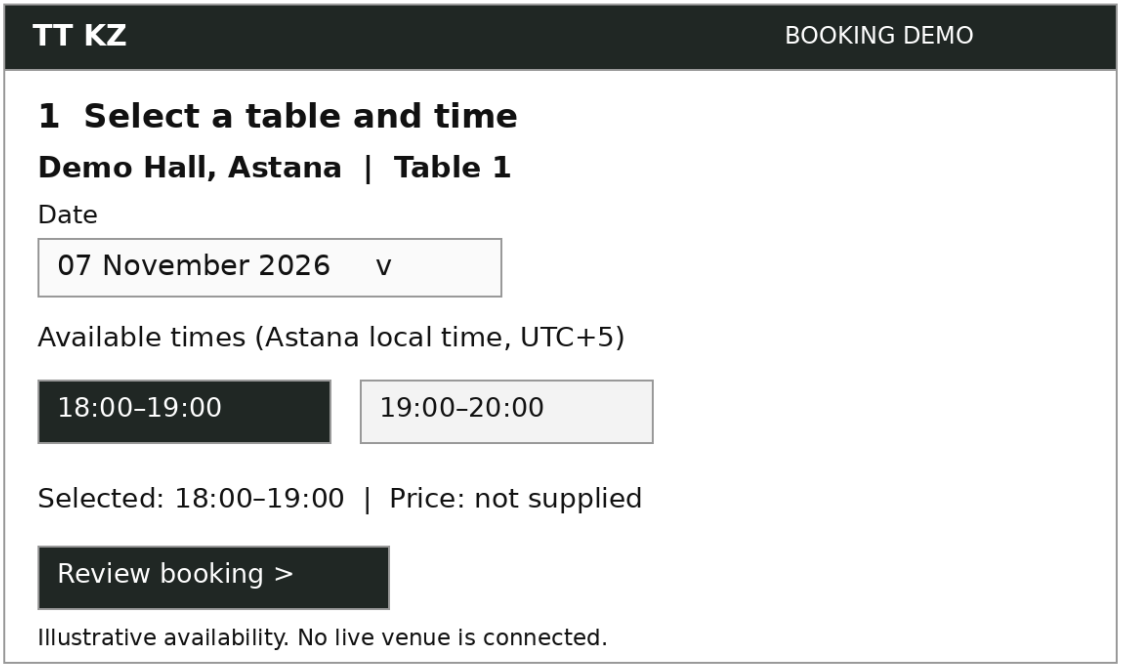

TTKZ — Sports halls
Спросить Gemini
127.0.0.1:5173/halls
TTKZ
Home
Sports halls
News & tournaments
Find players
About
Contact
Sign in
A TABLE FOR EVERY GAME
FIND YOUR PLACE TO PLAY.
Explore spaces to train, rally, and meet the community. Starting with Astana, built with Kazakhstan in mind.
Design catalog · Fictional halls and prices · Live availability is not yet connected
Search halls
Hall name or location
City
Astana
Facilities
Any facility
Up to 4,000 ₸ / hour
2 concept halls
Availability filter arrives with the booking API
Astana
Capital TT Arena
Demo venue
Astana
Saryarka Table Tennis
Demo venue

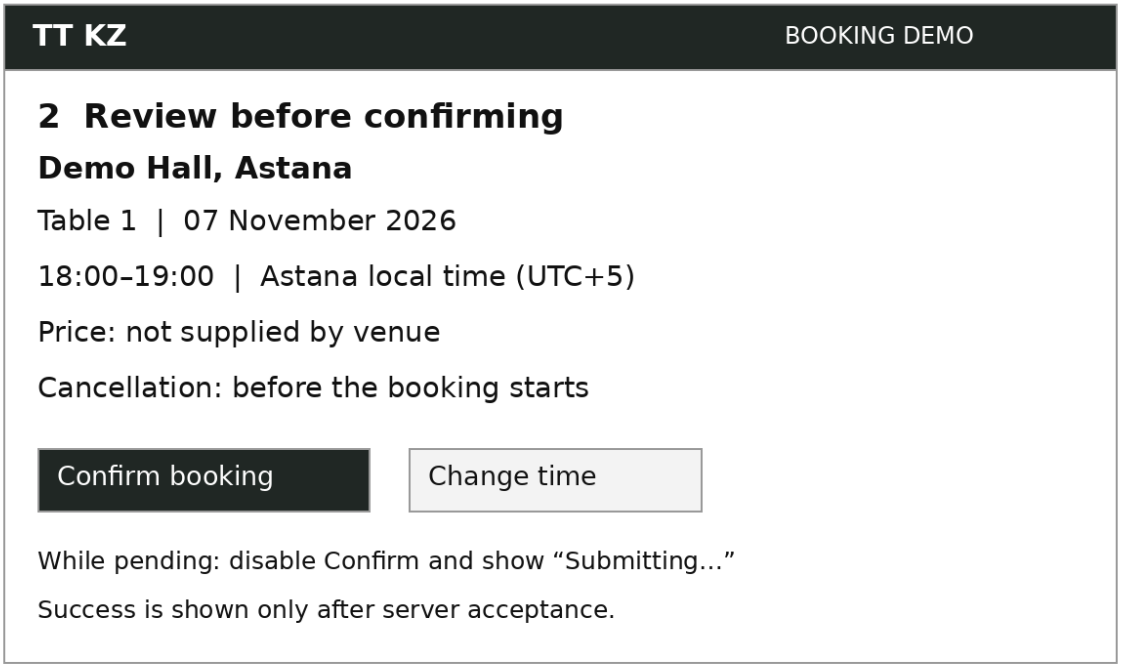

TTKZ — Your next game start
Спросить Gemini
127.0.0.1:5173
TTKZ
Home
Sports halls
News & tournaments
Find players
About
Contact
Sign in
• TABLE TENNIS. ALL TOGETHER.
PLAY MORE.
COMPETE BETTER.
CONNECT.
Your table. Your people. Your next great game.
A new home for Kazakhstan's table tennis community.
Book a Hall
Find Players
From Astana. For every player in Kazakhstan.
EXPLORE THE PLATFORM
ONE TABLE. ENDLESS POSSIBILITIES.
01 / 04

8.1 Booking flow and interaction states

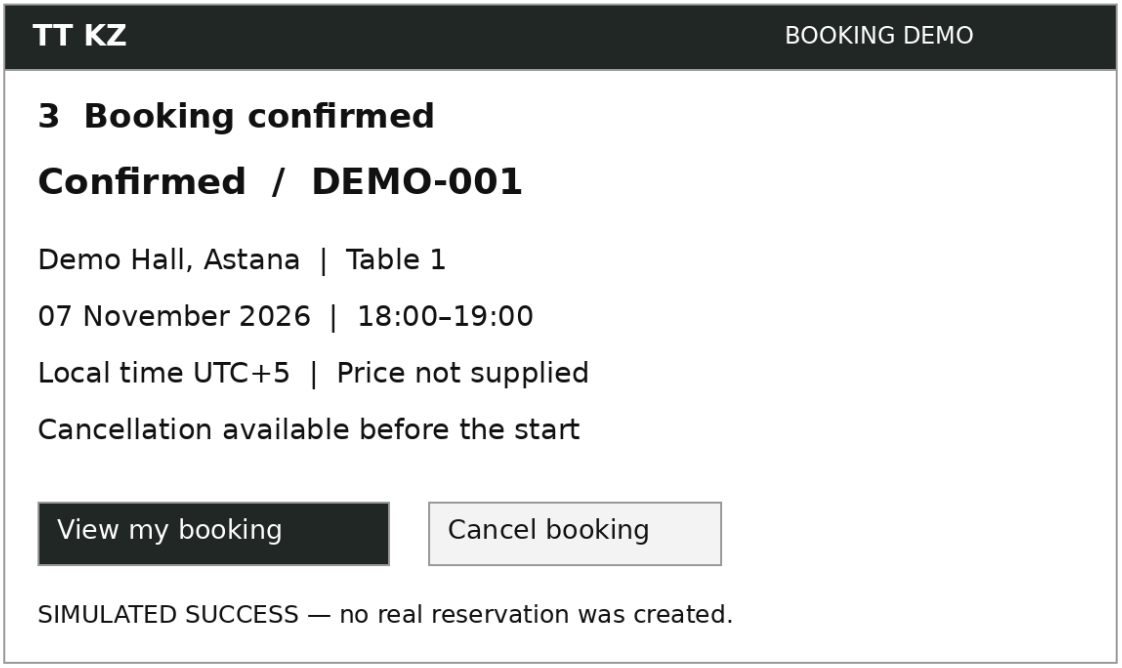
The following illustrative mockups describe time selection, review, confirmation and conflict recovery. Dates, venue details and booking IDs are demonstration data. The conflict image is a design proposal, not a screenshot of the current frontend. No database or API is connected and no actual reservation is evidenced.

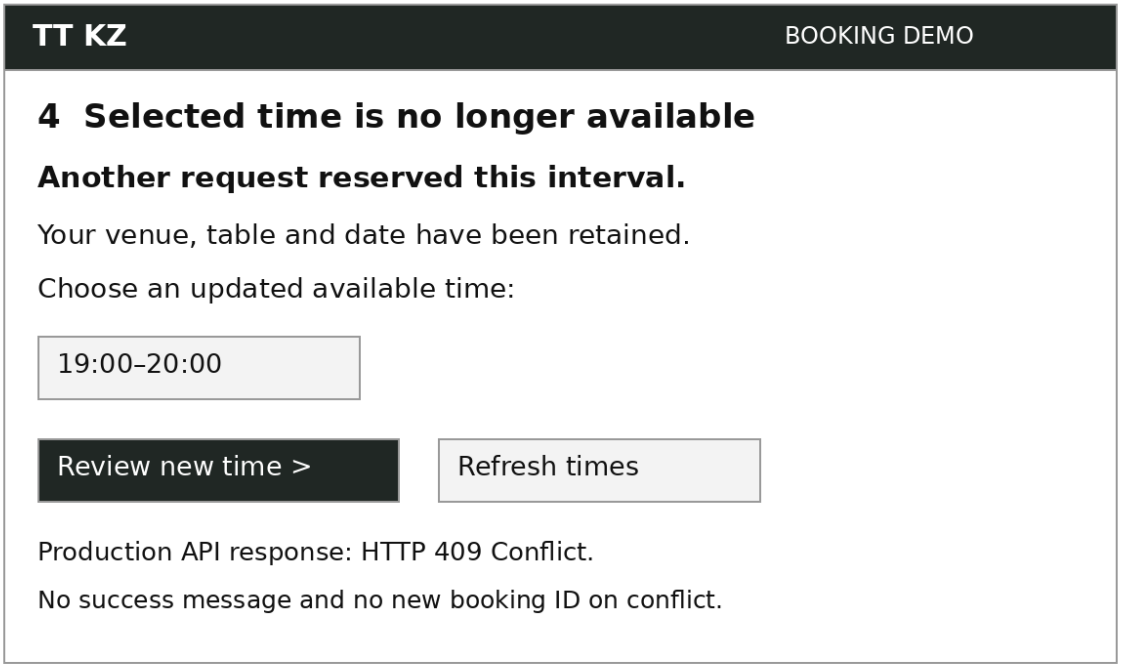
Time selection and review before submission





Successful confirmation and conflict recovery





Planned booking behavior: the confirmation action will be disabled while a request is pending. A booking ID will be displayed only after server acceptance. On HTTP 409, the interface will retain the venue and date, refresh available intervals and ask the user to select another time. Network failures will not be displayed as successful bookings. These behaviors are proposed and have not been tested against a connected backend.

Editable mockup sources: docs/prototype/booking-selection-review.svg and docs/prototype/booking-success-conflict.svg.

9. Verification and evaluation plan

Quality objective	**Method**	**Metric / baseline**
No conflicting reservations	Issue simultaneous requests to one table/time across a test database	Count overlaps: target 0; compare to intentionally unprotected availability-check flow or manually managed process.
Functional correctness	Automated API tests and manual acceptance checks mapped to FR-01–FR-11	Pass rate per requirement; failing cases and defect records.
Usability	Short moderated task evaluation with potential users	Success %, task duration, misclicks or errors, qualitative feedback; compare first versus revised prototype.
API behavior	Timed requests in an identical configured environment	p50/p95 latency and error rate; baseline from initial prototype or no-index version.
Access control	Negative tests across player and manager roles	No unauthorized reading/modification of protected records.

Proposed tools: pytest and HTTP testing for FastAPI, PostgreSQL integration test database for transaction tests, browser-based user-task observation for the frontend. Measurements must report machine/configuration, test date, sample size, number of concurrent users, and limitations. No results for the evaluations in this section have been supplied. Historical frontend build/check records in docs/VERIFICATION.md remain separate; they do not demonstrate backend correctness.

> All thresholds below are proposed for the next implementation evaluation. They must be reviewed before testing; results are not yet available.

Evaluation	Planned conditions	Pass rule and baseline
Booking integrity	20 simultaneous clients request the same table and interval; repeat 100 rounds on a reset test database. Include adjacent slots, cancellations and different tables.	Exactly one confirmation per contested interval; zero overlaps. Compare with a deliberately unprotected check-then-insert flow in an isolated test environment only.
Usability	Five potential users perform hall discovery and booking without help on the same desktop setup. Record completion, time and errors; repeat after revisions.	At least 4 of 5 complete the flow; report median task time. Compare first and revised prototype; acknowledge small sample and learning effects.
Performance	20 concurrent users; 1-minute warm-up followed by 5 minutes; seed 100 venues and 10000 bookings. Document hardware, software, dataset and request mix.	Read endpoint p95 <= 1000 ms; unexpected errors < 1%. Compare initial and revised implementation on the same environment; expected HTTP 409 conflicts are reported separately.
Functional and access control	Map tests to every Must FR/NFR; test unauthenticated, owner, other player and manager roles.	All Must acceptance checks pass; zero unauthorized reads or writes. Preserve failures and rerun evidence.

9.1 Operational outcomes for the Astana pilot

Outcome	Planned baseline and procedure	Proposed target
Booking effort	Five adults complete equivalent booking tasks using a prepared phone/message workflow and the TT KZ prototype. Counterbalance task order; measure elapsed time and user actions using the same venue data. Separate simulated task time from real venue response delays.	At least 20% lower median task time than baseline; at least 4 of 5 users complete without assistance. Report failures and small-sample limits.

Outcome	Planned baseline and procedure	Proposed target
Reservation conflicts	20 concurrent requests for the same table/interval, 100 repeated rounds; include adjacent intervals and cancellations. Compare to unprotected check-then-insert in an isolated test database.	0 overlapping active bookings; exactly 1 confirmation in each contested round; expected conflicts reported separately.
Access to relevant information	Users locate venue price/conditions and tournament date/location/requirements in equivalent prepared multi-channel information and TT KZ. Measure correctness, time and actions.	At least 4 of 5 users complete correctly; at least 20% lower median lookup time. Report unsuccessful tasks.
Data linkage and freshness	Create a manager availability update and organizer event update; check affected public view and cross-record IDs with seeded data.	All Must traceability cases pass; source and update time visible; unauthorized edits denied.

These are proposed thresholds, not achieved results. No attendance increase or nationwide demand will be inferred. The sample is for formative evaluation, not population estimates. The production booking-integrity test is deferred until the backend exists; current mockups cannot demonstrate database correctness. A01 supports preparing a phone/WhatsApp baseline, but administrator task time and actual conflict frequency were not measured. The planned five usability participants are separate from the five stakeholder interviews; recruitment overlap is not assumed.

10. Project management and risks

10.1 GitHub and YouTrack workflow

Repository: https://github.com/TTKZ12/TTKZ_diploma_work.

YouTrack: https://ttkz.youtrack.cloud/agiles/204-1/current?a=204-0%2Fcurrent

Workstream	Primary owner	Evidence to upload
Backend, DB and booking correctness	Zhanybek Alikhan	DB design, FastAPI contract, migrations, API tests, PRs
Frontend and flows	Tazhimova Araizhan	Layout screenshots, React components, user flows, tests, PRs

Workstream	Primary owner	Evidence to upload
User relevance	Tazhimova Araizhan	Interview protocol and anonymized evidence
Literature and systems comparison	Zhanybek Alikhan	Cited report, comparison evidence and revision history
Integration and submission	Tazhimova Araizhan & Zhanybek Alikhan	Component contracts, YouTrack links, reviewed tag

10.1.1 Version history and implementation evidence

The preserved probation tag prob-period-1 points to commit 81dbc95c9ef99a3c53024cb173e4833c60f7791f. The earlier repository inspection covered main-branch commit 40331ea7a2c4d4e47f0a00d341e7f21f7dcb18ee. These are historical snapshots. The revised documentation and supporting materials were subsequently published in commit 65da696ee8eb19280368cdb84276b64224c0d8ae. The original probation tag remains unchanged.

Revised materials: https://github.com/TTKZ12/TTKZ_diploma_work/commit/65da696ee8eb19280368cdb84276b64224c0d8ae

In the submitted tag and the earlier inspected main-branch snapshot, the backend directory contained only README.md. Those snapshots did not provide a runnable FastAPI application, database migrations or backend tests. The documentation publication described above does not itself demonstrate backend implementation. TTKZ-2 requires separate implementation evidence before it can be considered complete.

Student or account	Verifiable change	Version limitation
Tazhimova Araizhan – RII77777	Account identity confirmed by the team. Commit author labels RI and Arai T occur in the inspected history.	Team declaration establishes identity; profile/PR links should be added to README for reviewer verification.
Araizhan frontend change	3341b40ecd1e16f12d267c5f578a3812878f8d67: player filter reset; merged through PR #4.	Present in the earlier inspected main snapshot.
Araizhan branch changes	26a44a5: trim catalog queries; 29a307a: tournament clear-search button.	On frontend-improvements; not in the earlier inspected main snapshot.

Student or account	Verifiable change	Version limitation
Alikhan documentation	e4ac73b: backend planning status; dd1d17b: pre-defense README and supporting records.	Evidence of planning/documentation, not backend implementation.

The documentation revision is recorded separately from the earlier frontend contributions. Its publication does not establish that the backend or planned evaluation tests have been completed.

10.1.2 TTKZ-2 reconciliation and completion rules

The screenshot of TTKZ-2 explicitly requires backend implementation: FastAPI structure, PostgreSQL connection, SQLAlchemy models and Alembic migrations. Reopen it as In Progress if implementation is underway, or To do if it has not started, while these deliverables are absent. Retain the original scope and history, add a correction comment, assign the actual owner and link acceptance evidence. Do not rename an implementation task as planning merely to retain Done. No live status change is claimed here.

Work item	Owner and target	Acceptance evidence before Done
TTKZ-2 backend implementation	Alikhan; replan in YouTrack	Committed runnable FastAPI app, dependency/configuration examples, clean setup instructions, linked PR and relevant passing tests. Reopen while these are absent.
Backend schema and migrations	Alikhan; proposed 6 Nov 2026	User, venue, table and booking schema; migration from empty DB; overlap constraint and reviewed PR.
Booking API and authorization	Alikhan; proposed 14 Nov 2026	Create/cancel endpoints; ownership checks; conflict returns 409; concurrent integration tests.
Booking interface integration	Araizhan; proposed 21 Nov 2026	Time selection, pending, success and conflict states connected to API; error recovery and reviewed PR.
Probation evidence reconciliation	Both; before revised submission	Report, editable diagrams, provenance records, contribution links, review log and YouTrack statuses agree with the identified revision.

10.2 Risk register

Risk	Impact	Mitigation / owner
Lack of facility cooperation	Missing real schedule data	Start with an approved small pilot/demonstration dataset; identify partner early. Joint.
Double booking	Core correctness failure	Enforce database invariant; run concurrent API tests. Zhanybek.
Unclear API contract	Integration delays	Agree endpoint schemas and example payloads before UI wiring. Joint.
Scope expansion	Missed deadlines	Preserve five core features; defer payments, AI and federation integrations. Joint.
Insufficient interview evidence	Weak justification	Verify dates, formats, consent and original records for the five reported responses. Conduct new sessions only where necessary and record their actual dates. Joint.
Incomplete feedback record	Submission risk	Retain the received committee feedback forwarded by the supervisor, confirm its original date and document responses and outstanding actions. Request review of the revised materials. Joint.

10.3 Proposed milestones

The dates below are team planning targets, not university-approved deadlines. The first submission target is 24 October 2026. Later targets must be confirmed with the supervisor.

Target date	Deliverable and owner	Dependency and exit criterion
24 Oct 2026	Pre-defense evidence package; both students	Initial deadline: four real stakeholder exchanges. A01 obtained 25 Oct; committee feedback forwarded by the supervisor reported 29 Oct. These are later additions, not evidence available by 24 Oct.
6 Nov 2026	Database schema, migrations and API contract; Alikhan, reviewed by Araizhan	Agreed MVP; migration from empty database, reviewed schema, endpoint definitions and example payloads.
14 Nov 2026	Booking and authentication API; Alikhan	Schema and contract; ownership and concurrency acceptance tests pass.
21 Nov 2026	Core interface flows; Araizhan	API contract; venue, booking, player and tournament flows cover success and error states.
28 Nov 2026	Integrated MVP; both students	Backend and frontend ready; all Must acceptance tests pass.
5 Dec 2026	Pilot evaluation and revisions; both students	Integrated MVP; usability, load and integrity results documented; defects prioritized.

Task traceability: each meaningful YouTrack task records owner, priority, deadline, expected result and acceptance criteria. Include its ID in the branch and commit or PR description; link the reviewed PR back to the task. Close only after acceptance. Each student uses an identifiable personal account.

11. Committee feedback, supervisor review and submission readiness

11.1 Committee decision and communication

The probation committee allowed the team to continue work on the selected diploma topic and identified issues requiring correction. The supervisor forwarded the committee's feedback to Tazhimova Araizhan and requested that the comments be addressed.

The team reports receiving the forwarded feedback on 29 October 2026. The screenshots display a weekday rather than a full calendar date. This is the reported receipt date, not an independently verified date of the committee's review.

The committee's comments and the team's responses are recorded below and in Appendix F. Separate dated feedback from the supervisor on the revised materials remains to be obtained.

11.2 Responses to committee comments

Committee comment	Response and location	Status
RII77777 contribution could not be evaluated	Account mapped to Araizhan; specific diffs and main/branch/tag distinctions added in 10.1.1.	Identity clarified; add profile and task/PR links externally.
TTKZ-2 Done does not match committed backend	Implementation status corrected and completion criteria provided in 10.1.2.	Report corrected; live YouTrack correction pending.
Interview dates, formats and originals unverified	Four Telegram exchanges dated 24 Oct and A01 WhatsApp dated 25 Oct recorded; illustrative A01 removed and replaced by actual-response summary.	Content corrected; original screenshots and current translation checks pending.
TTFRK comparison unverified	Official home, calendar, rating and team pages checked on 29 Oct; functions and limits documented in 4.	Public-page verification completed.
Nine papers versus eleven rows; missing references	13 papers reconciled across narrative, matrix and bibliography, including Dašić and Wan.	Corrected in this revision.
Provide separate dated supervisor feedback with documented team responses.	Forwarded committee feedback is retained in Appendix F.1. Separate supervisor feedback will be recorded in Appendix F.2, together with the team's actions and evidence.	Pending: separate dated supervisor feedback has not yet been provided.

Committee comment	Response and location	Status
Define digital integration and geographic pilot	Participants, shared records and workflows specified; initial pilot restricted to Astana in 1.4.	Defined as proposed scope; venue partnerships not claimed.
Define measurable outcomes	Booking effort, conflicts, information access and data linkage mapped to methods and targets in 9.1.	Evaluation plan updated; results deferred.
Add report and editable diagrams to repository	Revised report plus editable draw.io diagrams and mockups prepared in companion package.	Files prepared; remote upload/PR not performed.
Booking prototype lacks time, confirmation and conflict states	Design mockups and expected transitions added in 8.1.	Selection/confirmation evidence reported; conflict remains a design mockup; DB/API not connected.

11.3 Evidence status and remaining actions

- Original conversation screenshots for all five participants are included in Appendix E. Interview dates, channels and source limitations are recorded in the provenance register.
- Screenshots of the committee feedback forwarded by the supervisor are included in Appendix F.1. The team reports receiving the message on 29 October 2026; the screenshots display a weekday rather than a full calendar date.
- Correct TTKZ-2 in YouTrack and connect tasks to the actual PRs, commits and acceptance criteria.
- Commit this revised report, editable diagrams and comparison register through a reviewed PR. Record the new revision URL without moving prob-period-1.
- Keep the backend labeled planned until committed code and checks meet the task acceptance criteria.
- Obtain separate dated feedback from the supervisor on the revised report and project materials. Record each comment, the team’s response and the supporting section or repository link in Appendix F.2.

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Appendix A Interview protocols and evidence fields

A.1 Session record

For each participant, retain the actual session date, written channel, role, interviewer, Russian source text, English translation, consent basis and source path. Tazhimova Araizhan conducted the interviews and Zhanybek Alikhan translated the responses. The team reports agreement to anonymized use; no signed consent form or screenshot proof of that agreement is claimed. Record actual dates for checking the current English text against original messages. A01 cannot have been checked on 24 October because the responses were received on 25 October.

A.2 Player questions for P01 to P03

1. How often do you play table tennis?
2. How do you usually find a sports hall or table to play at?
3. Describe how you arranged your most recent playing session or table reservation.
4. What difficulties have you experienced when looking for an available time slot or venue?
5. How do you find opponents with a similar playing level?
6. Where do you find information about amateur table tennis tournaments?
7. What information about a sports hall is important to you before visiting?
8. Do you currently use any websites or applications to organize table tennis games? If so, which ones?

A.3 Hall administrator questions for A01

Opening request: “Could you please answer the following questions?” This is an invitation, not a seventh research question. The six substantive questions follow.

1. How are bookings currently made: by phone, WhatsApp, website, or in person?
2. How is the table availability schedule managed?
3. Have there been cases of overlapping bookings? How were they resolved?
4. What questions do visitors ask most often?
5. Which scheduling tasks take the most time?
6. What limitations or difficulties might arise when using an online booking system?

A.4 Coach and tournament organizer questions for C01

1. How do you currently publish information about tournaments?
2. How do participants find out about tournament dates, locations, conditions, and skill levels?
3. What difficulties do you face when informing participants?
4. What digital tools do you already use?

5. What is missing from existing solutions?

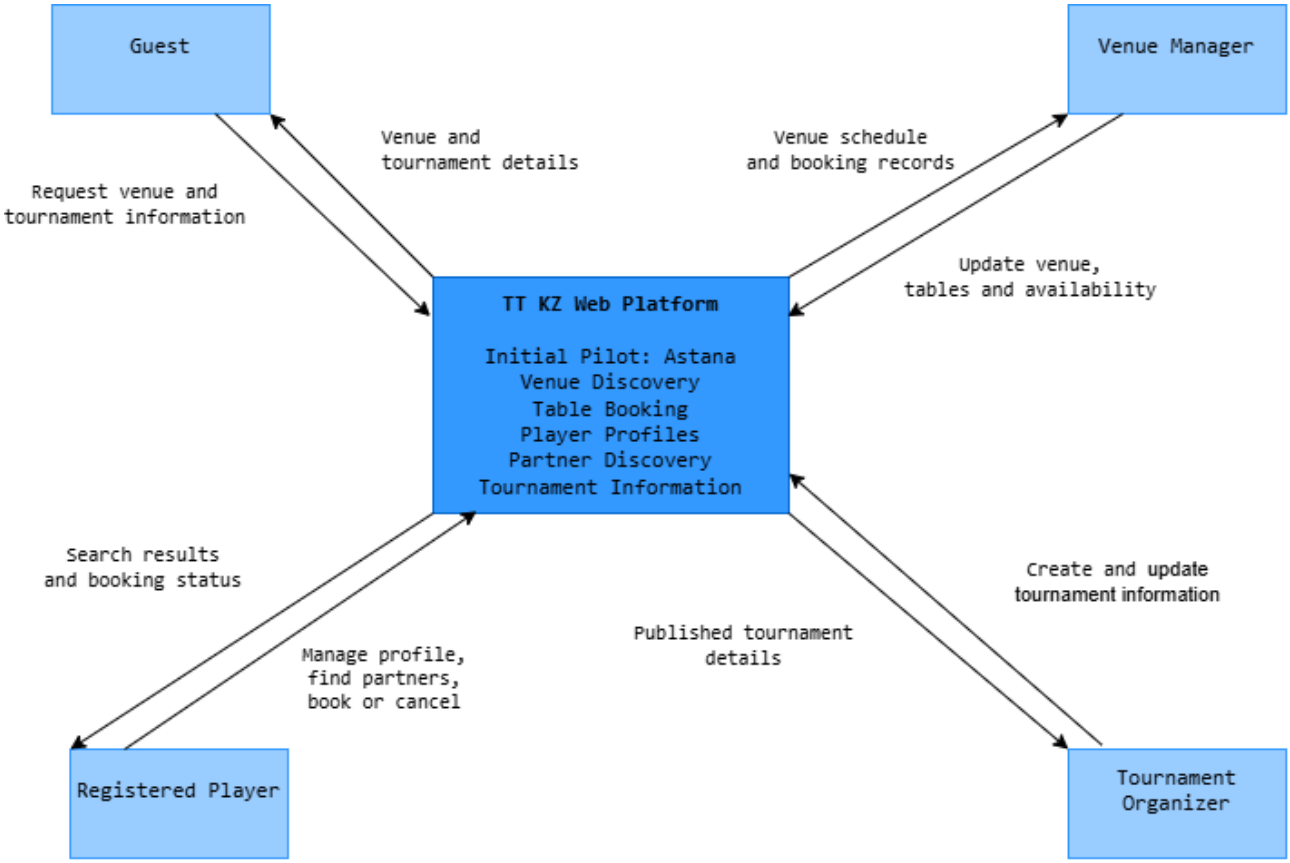
Appendix B. Artifact register

Artifact	Project file location	Status
Pre-defense report (DOCX)	docs/pre-defense/Prob_Period.docx	Editable master. Companion PDF: Prob_Period.pdf, exported from this same DOCX.
Interview evidence	docs/research/interviews/	Original screenshots for five participants are included in Appendix E. Responses were collected on 24-25 October 2026. The earlier illustrative A01 text has been replaced with actual responses.
Comparison evidence	docs/comparisons/screenshots/	Official-page screenshots and source links are included in Appendix G. Capture dates are recorded in the figure captions.
Context/use-case/component diagrams	docs/diagrams/	Three diagrams are included in Appendix C. Editable draw.io sources and readable exports are available in docs/diagrams/.
Prototype screenshots / Figma link	docs/prototype/	Frontend/demo screenshots; conflict mockup only, no conflict screen screenshot. No DB/API connection.
Test plan & results	docs/evaluation/	Draft plan; results pending
Committee feedback and supervisor review log	docs/supervisor-feedback/	Forwarded committee feedback and team responses are included in Appendix F.1. Separate dated supervisor feedback and responses remain pending in Appendix F.2.

Appendix C. Architecture Diagrams

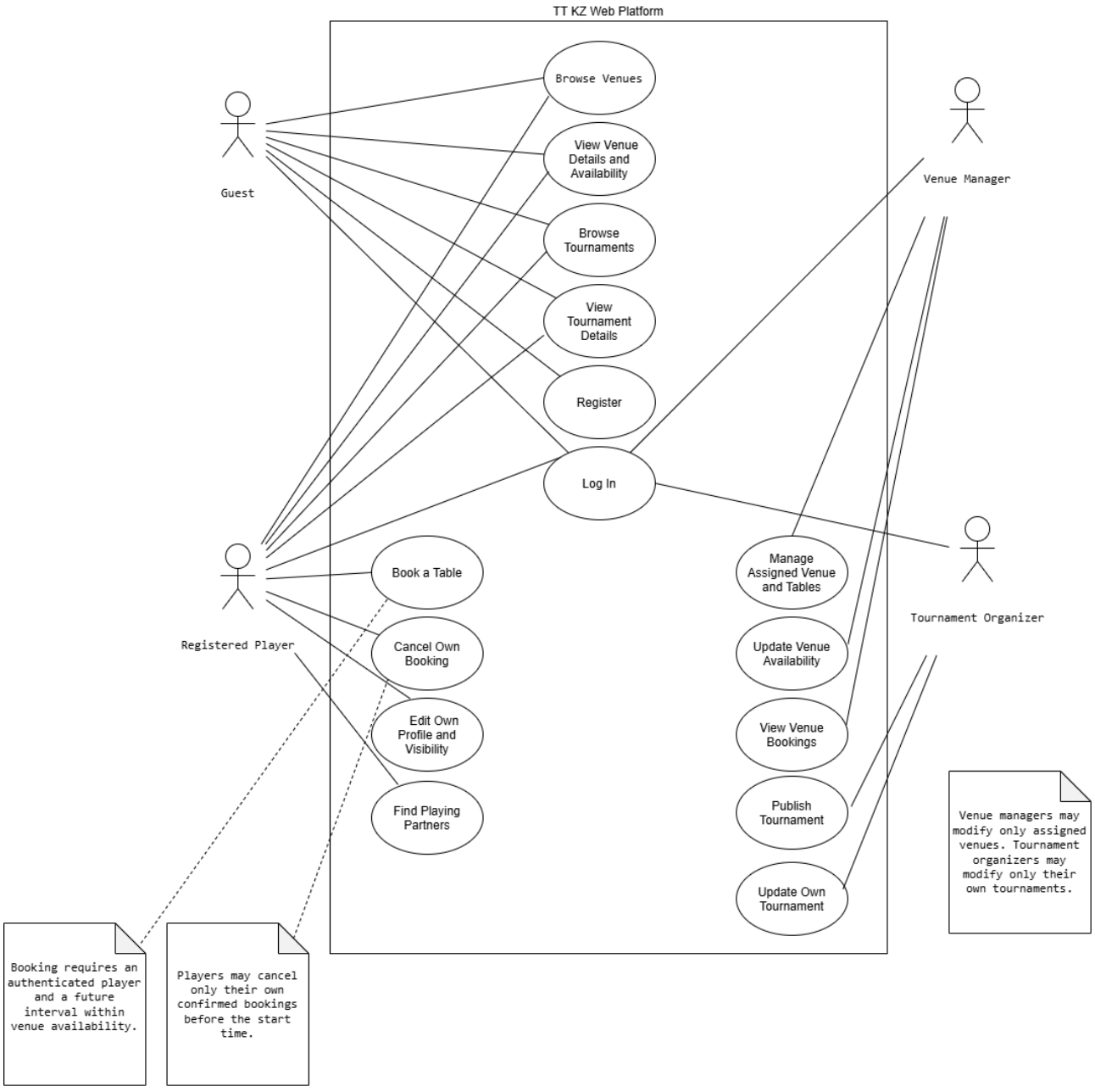
The following diagrams present the proposed TT KZ architecture for the initial pilot in Astana. Editable diagram sources are provided in the project repository.

C.1 System Context Diagram



The system boundary connects guests, players, venue managers and tournament publishers. No external federation API or messaging synchronization is assumed.

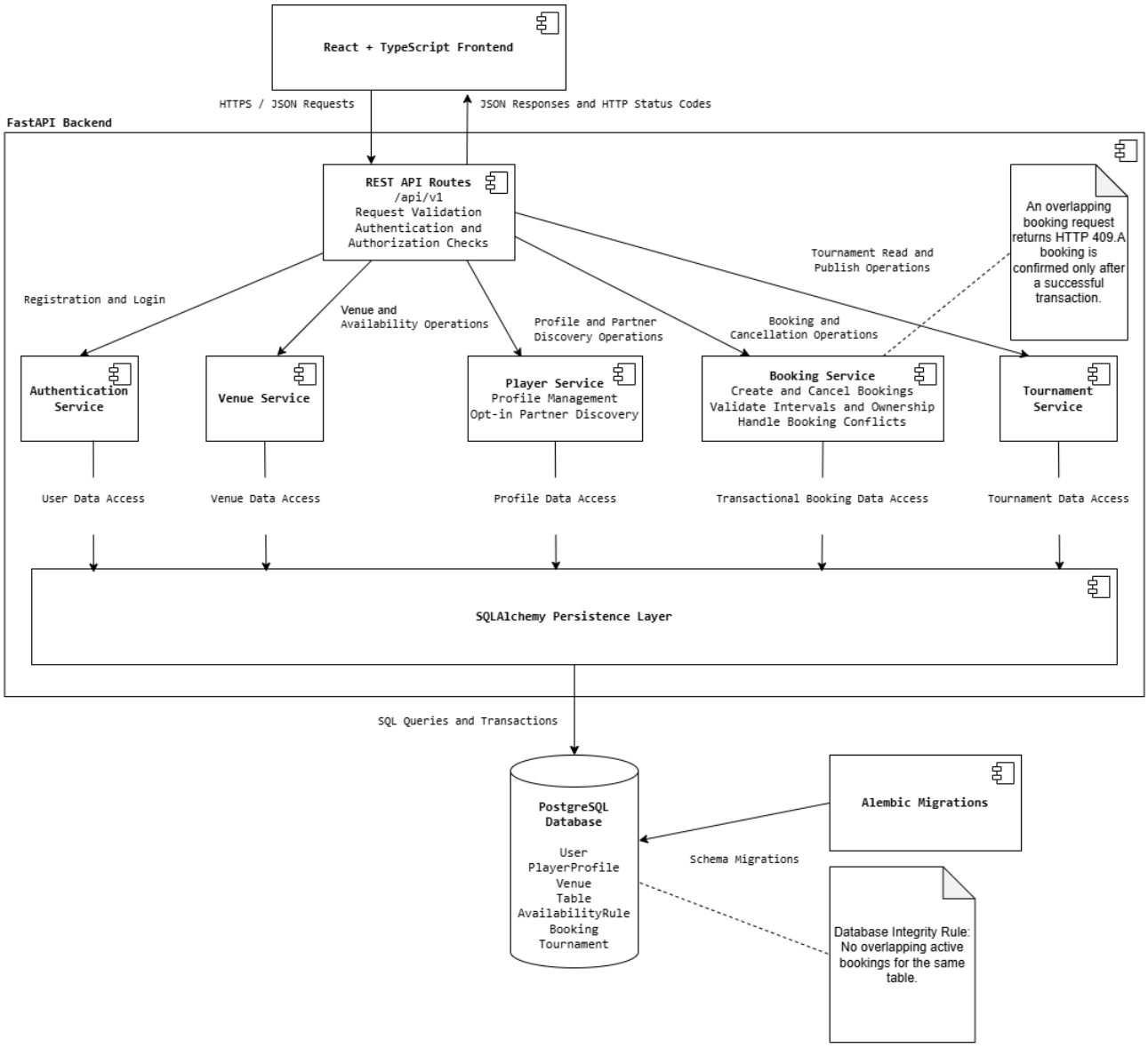
C.2 UML Use Case Diagram



Proposed MVP functionality. Implementation status is documented separately.

Actors connect to use cases inside the TT KZ boundary. Authentication and available time are booking preconditions. Server-side authorization and atomic non-overlap enforcement are required independently of these interface scenarios.

C.3 Component Diagram



Proposed architecture for the Astana pilot.
The frontend prototype exists; backend and database integration are planned.

Selected technology stack and planned component interfaces. React communicates with FastAPI REST routes. Routes dispatch to authentication, venue, player, booking and tournament services. The services use the shared SQLAlchemy persistence layer; PostgreSQL stores related records and enforces integrity constraints. These interfaces are proposed architecture, not evidence of implemented backend code. Alembic manages migrations.

C.4 Consistency and verification notes

- Player reservation requires an authenticated user and an available slot; server-side booking must prevent conflicting bookings using PostgreSQL transactions and enforced integrity constraints.

- Guests can browse public hall and tournament information without an account.
- Player matchmaking is initially filter-based (location, skill level, preferences, availability); no AI matchmaking is claimed.
- Tournament publishing and facility administration are proposed manager workflows, subject to actual stakeholder evidence and MVP prioritization.
- No payment gateway, chat, live streaming, federation integration or machine-learning service is assumed in the first release.

Appendix D Stakeholder responses

The revised dataset contains four Telegram exchanges dated 24 October 2026 (P01-P03 and C01) and one WhatsApp exchange dated 25 October 2026 (A01), according to the team. Original responses were in Russian; Zhanybek Alikhan prepared the English translations. Earlier illustrative administrator answers are excluded and replaced in D.5. The other four participants' response content is retained. Appendix E provides source screenshot slots; verification of the current English text remains to be completed.

1. How often do you play table tennis?

P01: I train approximately once a week, and I mainly play in tournaments.

P02: Every day.

P03: I play table tennis very frequently, approximately six times a week.

2. How do you usually find a sports hall or table to play at?

P01: I usually ask friends who also play table tennis. They recommend places where halls and tables are available. If I need to find a new venue, I ask in group chats, check maps, or use 2GIS.

P02: It is quite easy. The availability of facilities is good.

P03: I usually find a sports hall or table through the 2GIS application by searching for "table tennis."

3. Describe how you arranged your most recent playing session or table reservation.

P01: I message my friends in a group chat, and we agree on a time. If we need a table, I contact the sports hall by message or phone to check availability. There is no single place where I can see the schedule and make a reservation immediately.

P02: We agreed on a time with friends and then found a sports hall where we could play.

P03: I usually arrange a table reservation by phone.

4. What difficulties have you experienced when looking for an available time slot or venue?

P01: It is difficult to know whether a table is available at a particular time without messaging the facility or visiting it. Schedules are often outdated, and sometimes I have to wait for a response. In the evenings, especially after work, tables become occupied quickly, so finding an available time requires manual coordination.

P02: I have not experienced any difficulties. Everything has been fine.

P03: The main difficulties are related to financial limitations and reserving a table at a suitable time.

5. How do you find opponents with a similar playing level?

P01: Mainly through tournaments, where it is easier to understand each player's skill level. Outside tournaments, finding an appropriate opponent is difficult because you cannot always know how well someone plays beforehand. I usually look through friends and other players I know.

P02: Through tournaments. By playing against different participants, I can evaluate their skill levels.

P03: I find opponents of a similar level through dedicated table tennis group chats.

6. Where do you find information about amateur table tennis tournaments?

P01: I usually hear about tournaments from friends and other players, group chats, and occasionally the social media accounts of federations and clubs. I have not come across a single amateur tournament calendar that provides dates, skill levels, and participation requirements.

P02: Through RTTF and group chats.

P03: I usually find information about amateur tournaments through dedicated WhatsApp groups or Instagram accounts.

7. What information about a sports hall is important to you before visiting?

P01: The number and condition of the tables, floor surface, lighting, and available space around each table. I also consider the current schedule, prices, changing rooms and showers, directions to the venue, and parking availability.

P02: Available space, lighting, and table quality.

P03: Information about the tables and the skill levels of potential opponents.

8. Do you currently use any websites or applications to organize table tennis games? If so, which ones?

P01: I do not use any specialized table tennis applications. We organize everything through messaging applications such as WhatsApp and Telegram, group chats with friends, 2GIS, and the Instagram accounts of sports clubs.

P02: RTTF and TTRanking.

P03: Yes, I use Unreal Engine 5 to create graphical games.

D.5 Hall administrator responses

English summary of A01's written Russian WhatsApp responses, 25 October 2026 (date). Repeated messages are consolidated without adding findings. This is a paraphrase, not a verbatim quotation. Source screenshot slot: Appendix E, Figure E-A01; use sequential suffixes for multiple images.

1. How are bookings currently made: by phone, WhatsApp, website, or in person?

A01 summary: Bookings are made by telephone or WhatsApp. Customers contact the administrator to check availability and reserve a suitable time. Booking about a day ahead is recommended, but same-day bookings are possible when tables are available.

2. How is the table availability schedule managed?

A01 summary: Approximately eight tables are usually available for rent. The administrator personally checks which tables are available or reserved. Each request specifies the date, time and number of guests. A table is assigned to the customer after confirmation.

3. Have there been cases of overlapping bookings? How were they resolved?

A01 summary: Several customers may request tables for the same time. The administrator checks available tables before confirming. If all tables are occupied, another time or date is offered. Interpretation: this describes competing requests, not an established case of two confirmed bookings for the same table.

4. What questions do visitors ask most often?

A01 summary: Visitors ask about available tables, same-day bookings, the number of available tables, choosing a specific table, deposits, payment methods, cancellation and rescheduling. Payment is made at the visit; no advance booking payment is required.

5. Which scheduling tasks take the most time?

No separate substantive answer was provided. No finding about the most time-consuming scheduling tasks is inferred.

6. What limitations or difficulties might arise when using an online booking system?

No separate substantive answer was provided. No finding about staff training, interface preferences or limitations of online booking is inferred.

D.6 Coach and tournament organizer responses

1. How do you currently publish information about tournaments?

C01: We usually publish tournament information through WhatsApp groups, Instagram, and sometimes Telegram. We share announcements, posters, and registration details. However, the information is spread across different platforms.

2. How do participants find out about tournament dates, locations, conditions, and skill levels?

C01: Participants usually check social media posts or ask questions in group chats. They may also contact the organizer directly to clarify the tournament format, participation requirements, or registration deadline.

3. What difficulties do you face when informing participants?

C01: The main difficulty is reaching all interested players. Some participants may miss announcements, while others receive outdated information. We also have to answer the same questions repeatedly and send updates if the schedule or location changes.

4. What digital tools do you already use?

C01: We mainly use WhatsApp, Instagram, Google Forms, and spreadsheets. These tools help us communicate with participants, collect registrations, and organize tournament information, but they are not connected in one system.

5. What is missing from existing solutions?

C01: It would be useful to have one platform where players can find upcoming tournaments, check the date and location, view participation requirements, and register. Organizers should also be able

to update information easily and notify participants about changes. A filter by city and playing level would make it easier for players to find suitable tournaments.

Appendix E Interview provenance register

Updated 2 November 2026. The revised dataset includes two written Telegram exchanges and one written TikTok exchange from 24 October, two written WhatsApp exchanges from 24-25 October. Interviewer: Tazhimova Araizhan; original language: Russian; English translation: Zhanybek Alikhan. According to Araizhan, all participants agreed to anonymized use. Documentary evidence of the agreement and verification of the current English version are not claimed. The original four records were reportedly checked on 24 October; A01 record was reportedly checked on 25 October.

Participant	Actual date and format	Original source and consent	Verification
P01	24 October 2026; written Telegram exchange.	Slot E-P01. Agreement to anonymized use reported by Araizhan; documentary evidence unspecified. Screenshot to insert.	Original check reported 24 Oct; current English version check pending.
P02	24 October 2026; written Telegram exchange.	Slot E-P02. Agreement to anonymized use reported by Araizhan; documentary evidence unspecified. Screenshot to insert.	Original check reported 24 Oct; current English version check pending.
P03	24 October 2026; written TikTok exchange.	Slot E-P03. Agreement to anonymized use reported by Araizhan; documentary evidence unspecified. Screenshot to insert.	Original check reported 24 Oct; current English version check pending.
A01	25 October 2026; written WhatsApp exchange.	Slot E-A01. Agreement to anonymized use reported by Araizhan; documentary evidence unspecified. Screenshot to insert.	Original check reported 25 Oct; current English version check pending.

Participant	Actual date and format	Original source and consent	Verification
C01	24 October 2026; written WhatsApp exchange.	Slot E-C01. Agreement to anonymized use reported by Araizhan; documentary evidence unspecified. Screenshot to insert.	Original check reported 24 Oct; current English version check pending.

E-P01. Original written responses – Regular player

Session: 24 October 2026. Channel: Telegram. Participant code: P01.

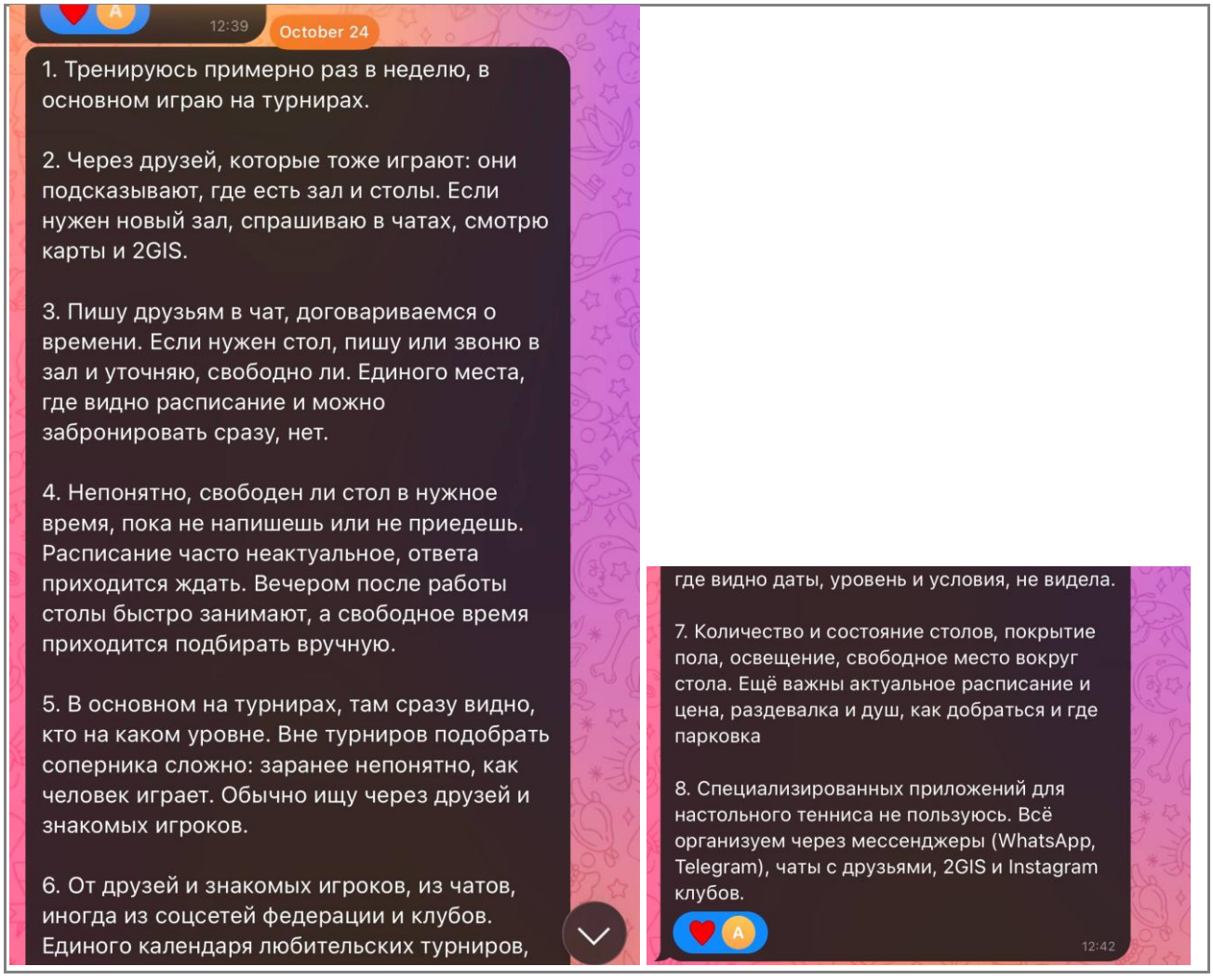


Figure E-P01. Anonymized written Telegram exchange, 24 October 2026. Source: docs/research/interviews/. Interviewer: Tazhimova Araizhan. Language: Russian. English translation: Zhanybek Alikhan. Original-response check reported by Araizhan: 24 October 2026.

E-P02. Original written responses – Regular player

Session: 24 October 2026. Channel: Telegram. Participant code: P02.

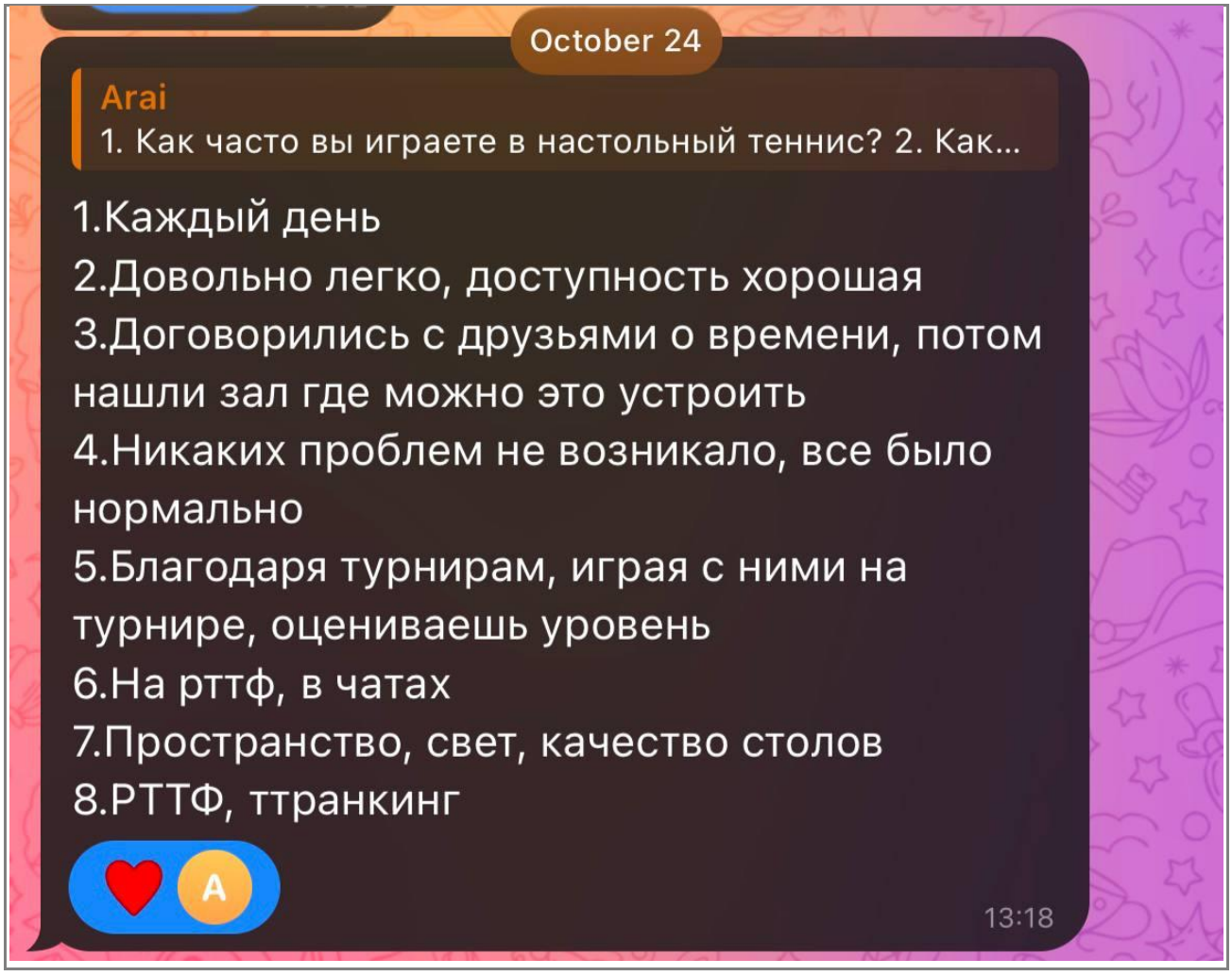


Figure E-P02. Anonymized written Telegram exchange, 24 October 2026. Source: docs/research/interviews/. Interviewer: Tazhimova Araizhan. Language: Russian. English translation: Zhanybek Alikhan. Original-response check reported by Araizhan: 24 October 2026.

E-P03. Original written responses – Regular player

Session: 24 October 2026. Channel: TikTok. Participant code: P03.

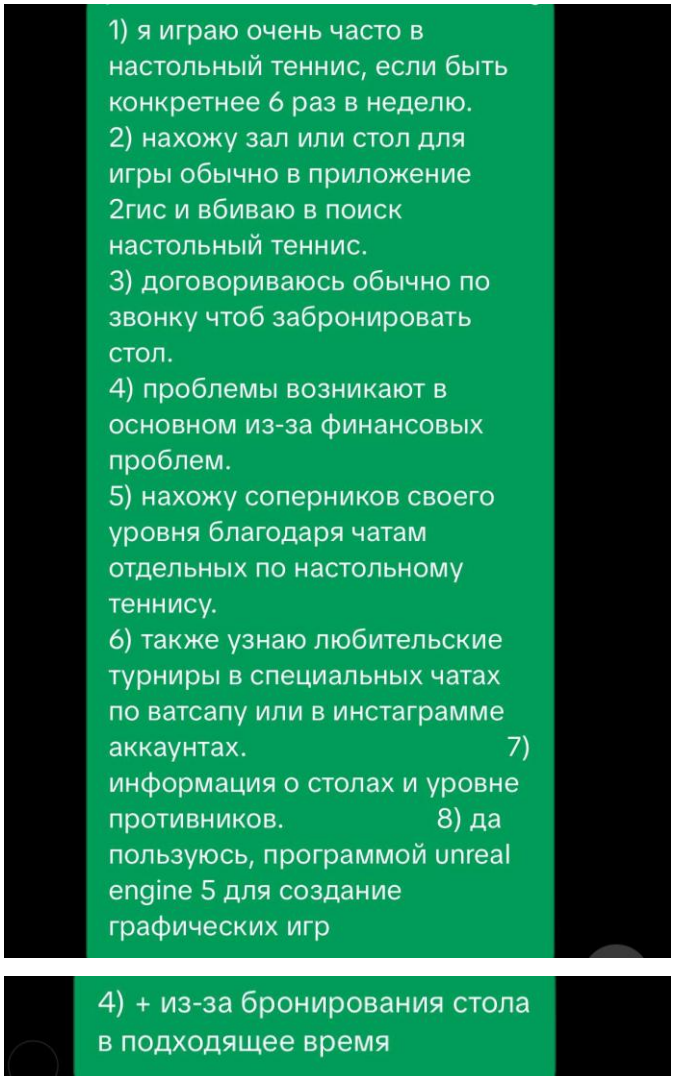


Figure E-P03. Anonymized written TikTok exchange, 24 October 2026. Source: docs/research/interviews/. Interviewer: Tazhimova Araizhan. Language: Russian. English translation: Zhanybek Alikhan. Original-response check reported by Araizhan: 24 October 2026.

E-A01. Original written responses – Hall administrator

Session: 25 October 2026. Channel: WhatsApp. Participant code: A01.

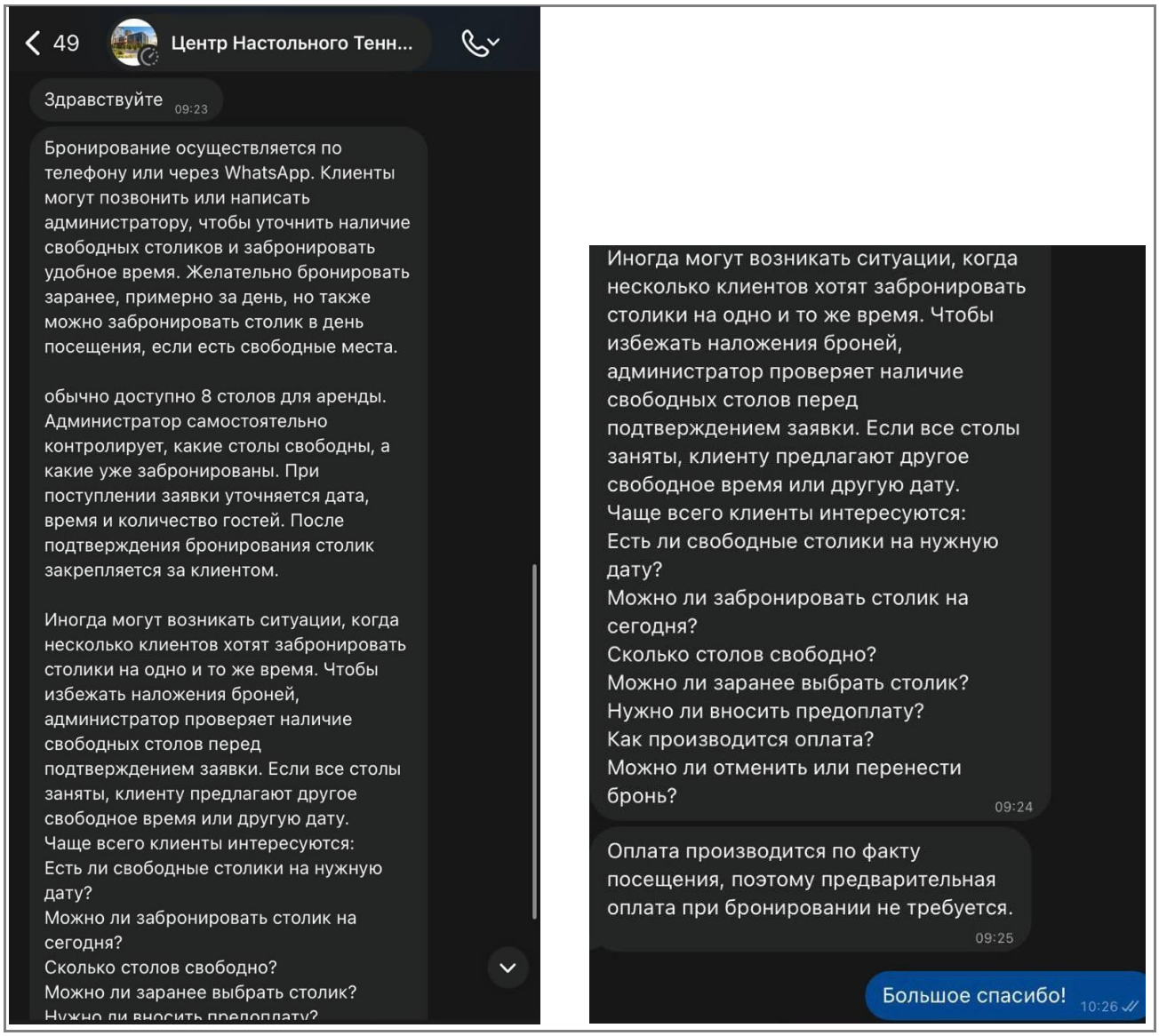


Figure E-A01. Anonymized written WhatsApp exchange, 25 October 2026. Source: docs/research/interviews/. Interviewer: Tazhimova Araizhan. Language: Russian. English translation: Zhanybek Alikhan.

E-C01. Original written responses – Coach and tournament organizer

Session: 24 October 2026. Channel: WhatsApp. Participant code: C01.

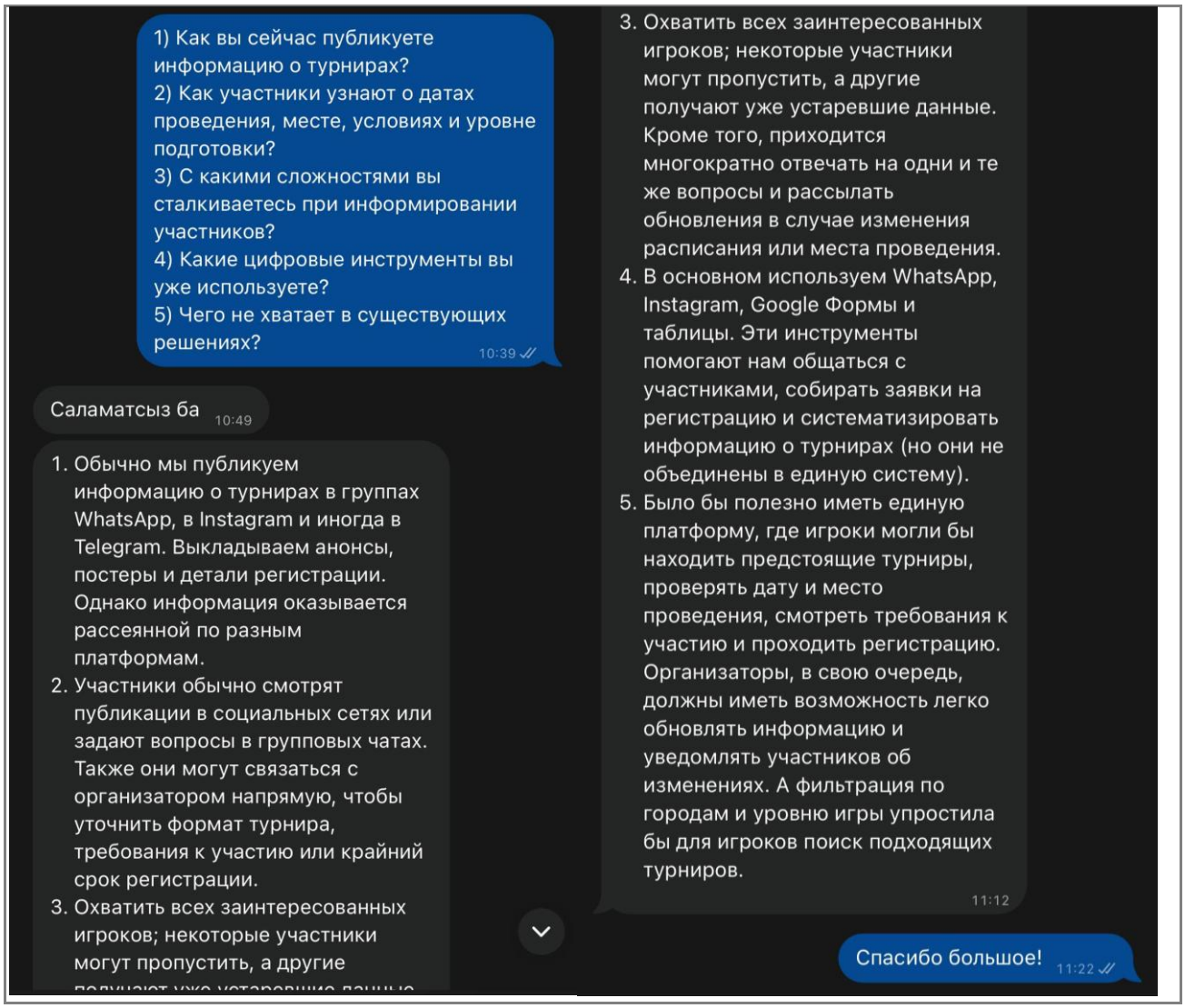


Figure E-C01. Anonymized written WhatsApp exchange, 24 October 2026. Source: docs/research/interviews/. Interviewer: Tazhimova Araizhan. Language: Russian. English translation: Zhanybek Alikhan. Original-response check reported by Araizhan: 24 October 2026.

Evidence for this study consists of written message exchanges and screenshots. No audio or video interview recording was made.

Appendix F Committee feedback and and supervisor review record

The following feedback was issued by the probation committee and forwarded to the team by the supervisor. Tazhimova Araizhan received the message. The team reports receipt on 29 October 2026; the supplied screenshots display the date. The forwarded comments are retained as committee feedback and do not constitute a separate supervisor review.

The submitted report and frontend prototype provide a sufficient preliminary basis for continuing the diploma project. The materials include a defined scope, literature and technology reviews, requirements with acceptance criteria, architectural diagrams, and a measurable evaluation plan. However, several probationary-period requirements remain insufficiently evidenced.

It is not possible to evaluate the contribution of the user with account RII77777.

The state of the repository does not match YouTrack. The task TTKZ-2, regarding the creation of the backend, is marked as Done, yet in the committed version, the backend directory contains only a README file. The outcome of this task needs to be clarified, and the statuses, links to changes, and acceptance criteria must be updated accordingly.

The responses of five participants are provided, but the dates, the format of the sessions, and the original interview recordings have not been verified. The comparison involving TTFRK remains unverified. The review mentions nine research papers, while the matrix lists eleven; Dašić and Wan are cited in the text but are missing from the bibliography.

Obtain the supervisor's feedback and include it in the report.

The topic has practical relevance, as it addresses the coordination needs of table tennis players, venue operators, and tournament organizers. The submitted stakeholder responses provide preliminary support for consolidating venue discovery, booking, partner search, and tournament information. However, the term digital integration requires a clear operational definition. The team should specify which participants, processes, and data the platform will connect, clarify the geographic scope of the initial pilot, and define measurable outcomes, such as reduced booking effort, fewer reservation conflicts, and improved access to relevant information. The current evidence does not establish nationwide demand.

The team must add the report and editable diagram sources to the repository, document interview provenance, resolve literature and reference inconsistencies, complete the comparison evidence, and provide dated committee feedback forwarded by the supervisor with documented responses. The booking prototype should cover time selection, confirmation, and conflict states. A complete application is not required at this stage, but the proposed integration must be reflected consistently in the requirements, architecture, prototype, and evaluation criteria.

Actual acknowledgement from Tazhimova Araizhan (Arai), shown at 12:24; message date not independently established: “Hello! Thank you for the feedback. I will review the comments and make the necessary corrections to the report and project”. The item-by-item response record is in section 11.

Additional supervisor instruction: “Please address the comments of the committee”.

Opening decision: “Hello! After consideration of your probation on diploma, you are allowed to continue your work on the selected topic”.

F.1 Committee feedback forwarded by the supervisor and acknowledgement screenshots

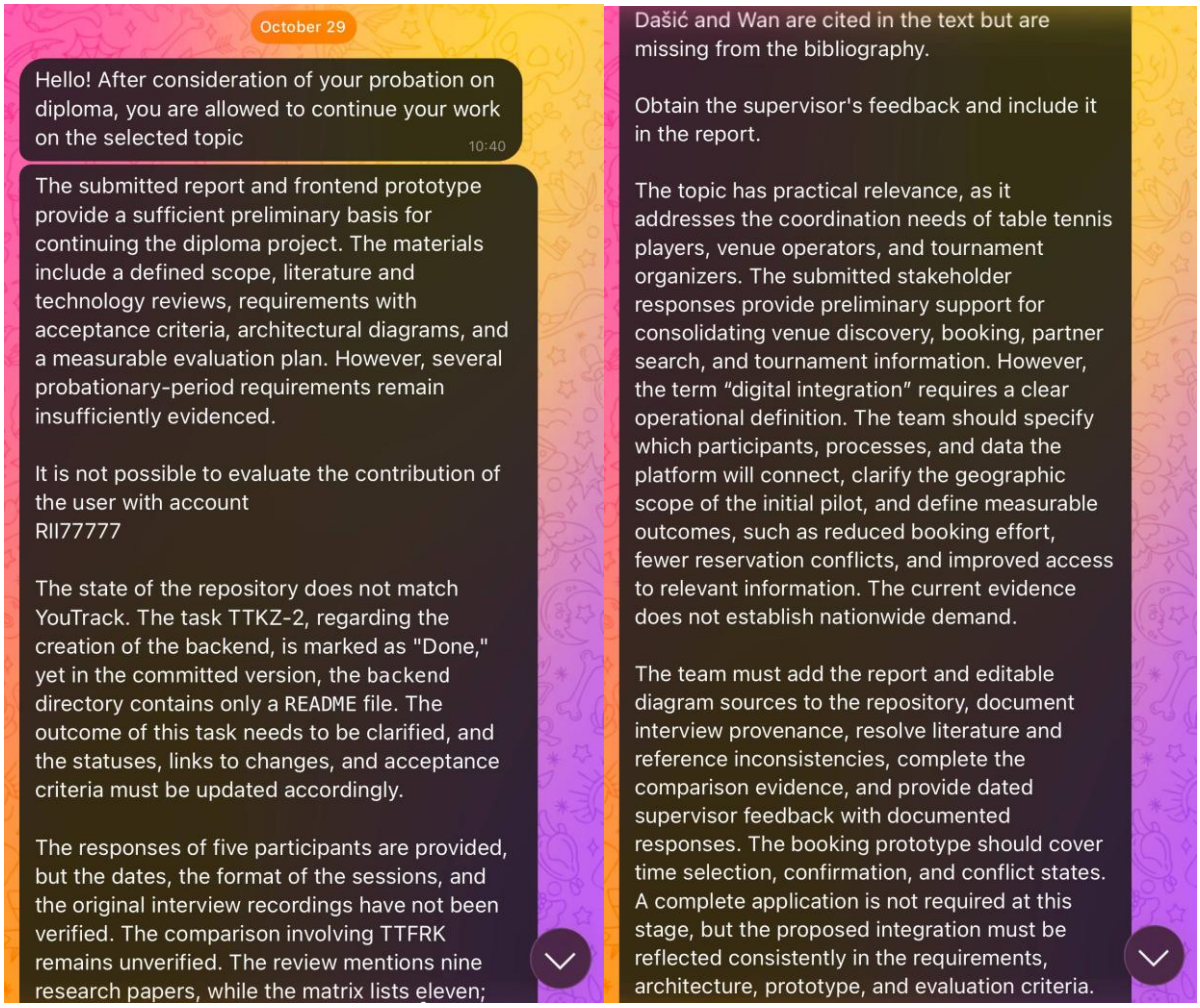


Figure F-01-02. Committee feedback forwarded by the supervisor received by Tazhimova Araizhan.

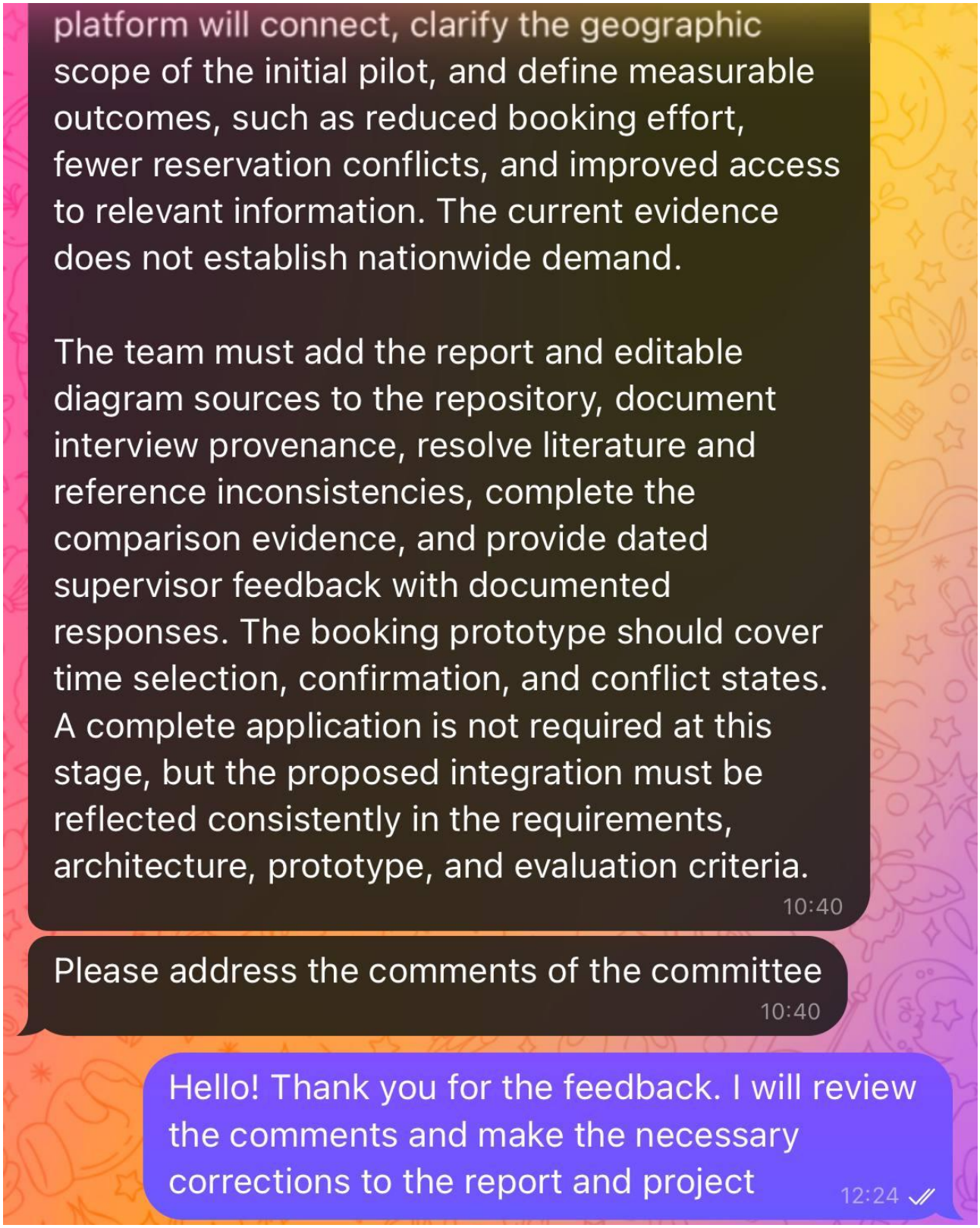


Figure F-03. Continuation of the forwarded committee feedback and related correspondence.

Tazhimova Araizhan acknowledged receipt and stated that the team would review the comments and make the necessary corrections. This acknowledgement is separate from the documented corrective actions in section 11.2.

Date: 29 October 2026.

Files: docs/supervisor-feedback/feedback-2026-10-29-01.png, feedback-2026-10-29-02.png, and feedback-2026-10-29-03.png

F.2 Separate dated supervisor feedback and team responses

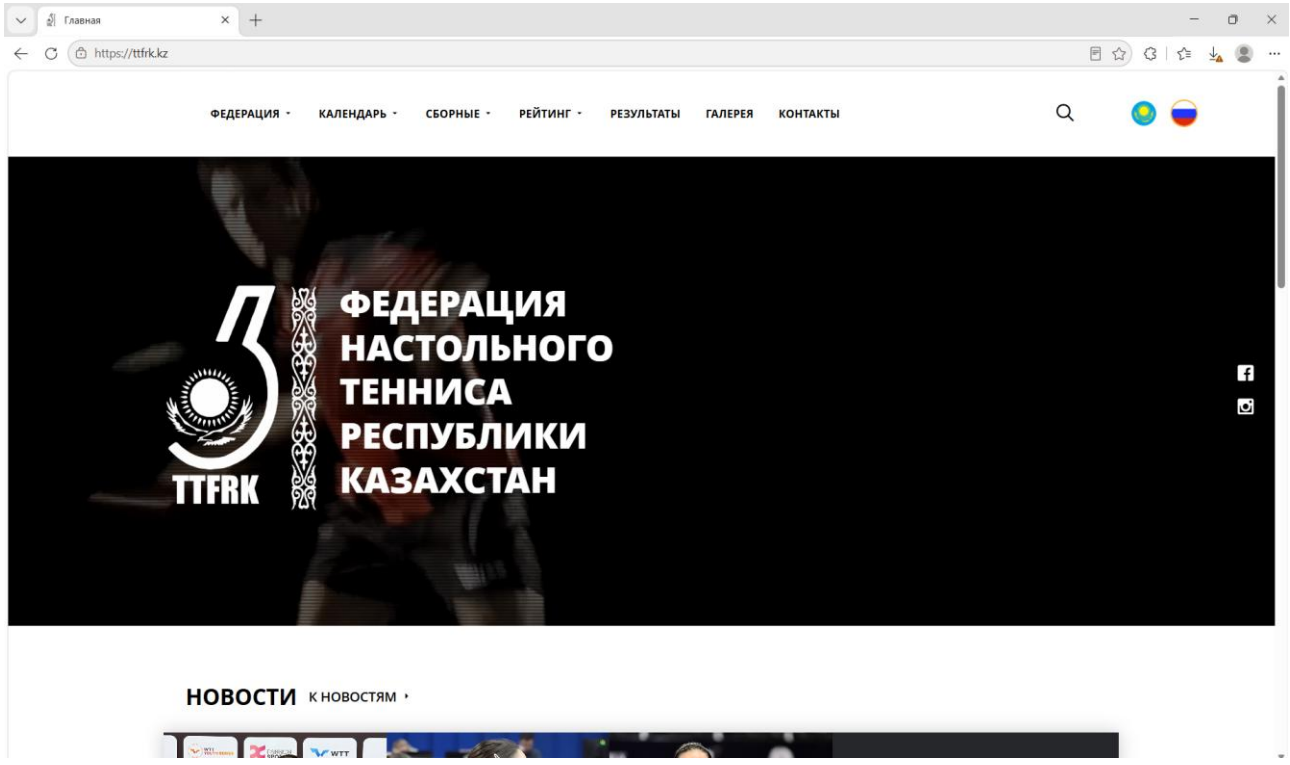
Separate dated feedback from the supervisor on the revised materials has not yet been received. This subsection will record the review date, the materials reviewed, the supervisor’s comments and the team’s responses with supporting evidence.

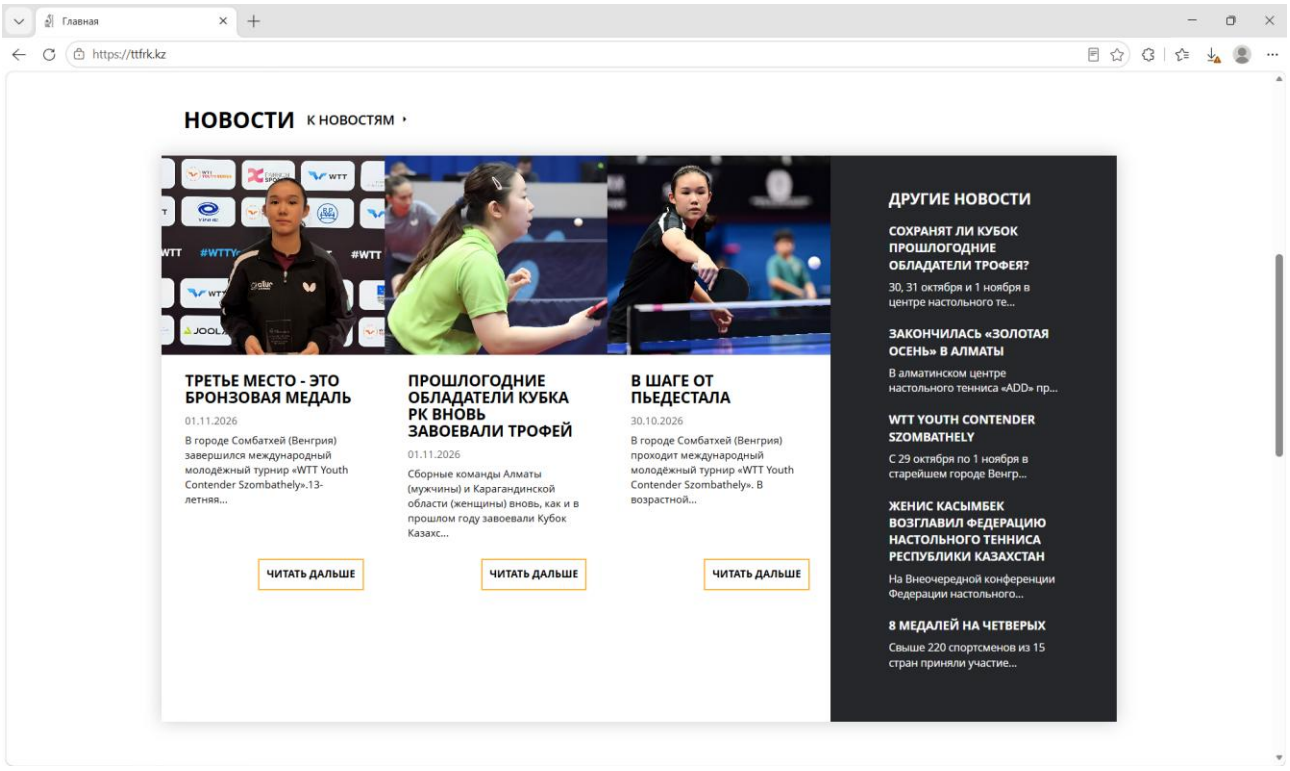
Appendix G TTFRK screenshot evidence

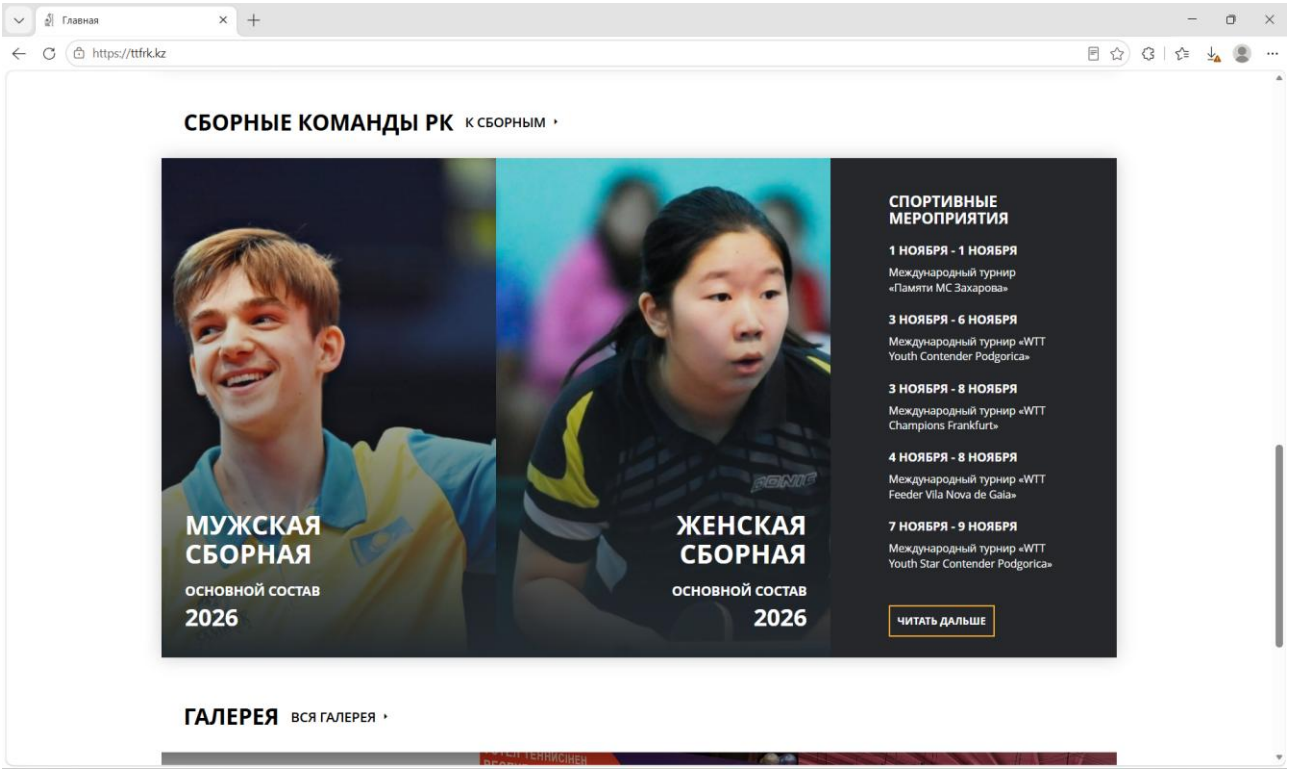
Official public pages rechecked on 29 October 2026. These pages support public-information observations; they do not prove the absence of booking or partner-search functions.

G-01-04. Official home

https://ttfrk.kz/




Главная
https://ttfrk.kz
НОВОСТИ К НОВОСТЯМ
ТРЕТЬЕ МЕСТО - ЭТО БРОНЗОВАЯ МЕДАЛЬ
01.11.2026
В городе Сомбатхей (Венгрия) завершился международный молодёжный турнир «WTT Youth Contender Szombathely».13-летняя...
ЧИТАТЬ ДАЛЬШЕ
ПРОШЛОГОДНИЕ ОБЛАДАТЕЛИ КУБКА РК ВНОВЬ ЗАВОЕВАЛИ ТРОФЕЙ
01.11.2026
Сборные команды Алматы (мужчины) и Карагандинской области (женщины) вновь, как и в прошлом году завоевали Кубок Казахс...
ЧИТАТЬ ДАЛЬШЕ
В ШАГЕ ОТ ПЬЕДЕСТАЛА
30.10.2026
В городе Сомбатхей (Венгрия) проходит международный молодёжный турнир «WTT Youth Contender Szombathely». В возрастной...
ЧИТАТЬ ДАЛЬШЕ
ДРУГИЕ НОВОСТИ
СОХРАНЯТ ЛИ КУБОК ПРОШЛОГОДНИЕ ОБЛАДАТЕЛИ ТРОФЕЯ?
30, 31 октября и 1 ноября в центре настольного те...
ЗАКОНЧИЛАСЬ «ЗОЛОТАЯ ОСЕНЬ» В АЛМАТЫ
В алматинском центре настольного тенниса «ADD» пр...
WTT YOUTH CONTENDER SZOMBATHELY
С 29 октября по 1 ноября в старейшем городе Венгр...
ЖЕНИС КАСЫМБЕК ВОЗГЛАВИЛ ФЕДЕРАЦИЮ НАСТОЛЬНОГО ТЕННИСА РЕСПУБЛИКИ КАЗАХСТАН
На Внеочередной конференции Федерации настольного...
8 МЕДАЛЕЙ НА ЧЕТВЕРЫХ
Свыше 220 спортсменов из 15 стран приняли участие...


Главная
https://ttfrk.kz
СБОРНЫЕ КОМАНДЫ РК К СБОРНЫМ
МУЖСКАЯ СБОРНАЯ
ОСНОВНОЙ СОСТАВ
2026
ЖЕНСКАЯ СБОРНАЯ
ОСНОВНОЙ СОСТАВ
2026
СПОРТИВНЫЕ МЕРОПРИЯТИЯ
1 НОЯБРЯ - 1 НОЯБРЯ
Международный турнир «Памяти МС Захарова»
3 НОЯБРЯ - 6 НОЯБРЯ
Международный турнир «WTT Youth Contender Podgorica»
3 НОЯБРЯ - 8 НОЯБРЯ
Международный турнир «WTT Champions Frankfurt»
4 НОЯБРЯ - 8 НОЯБРЯ
Международный турнир «WTT Feeder Vila Nova de Gaia»
7 НОЯБРЯ - 9 НОЯБРЯ
Международный турнир «WTT Youth Star Contender Podgorica»
ЧИТАТЬ ДАЛЬШЕ
ГАЛЕРЕЯ ВСЯ ГАЛЕРЕЯ

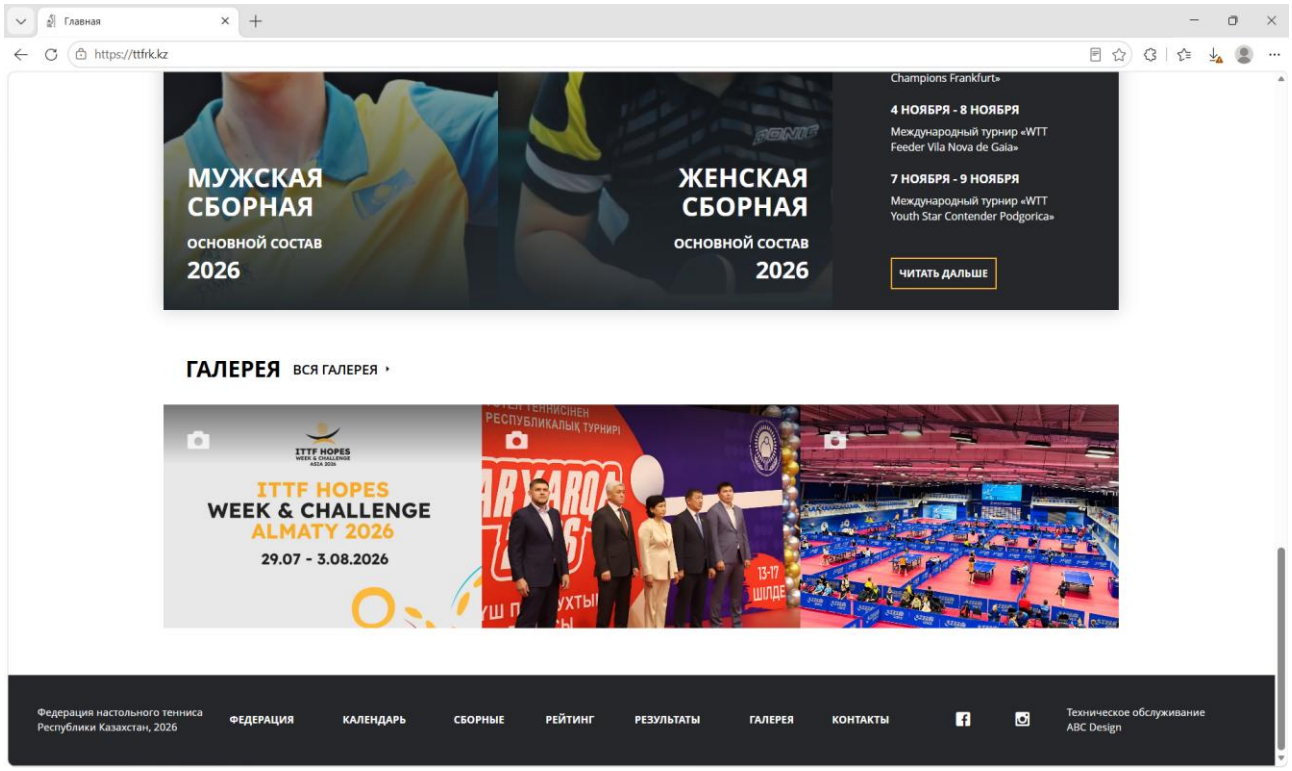
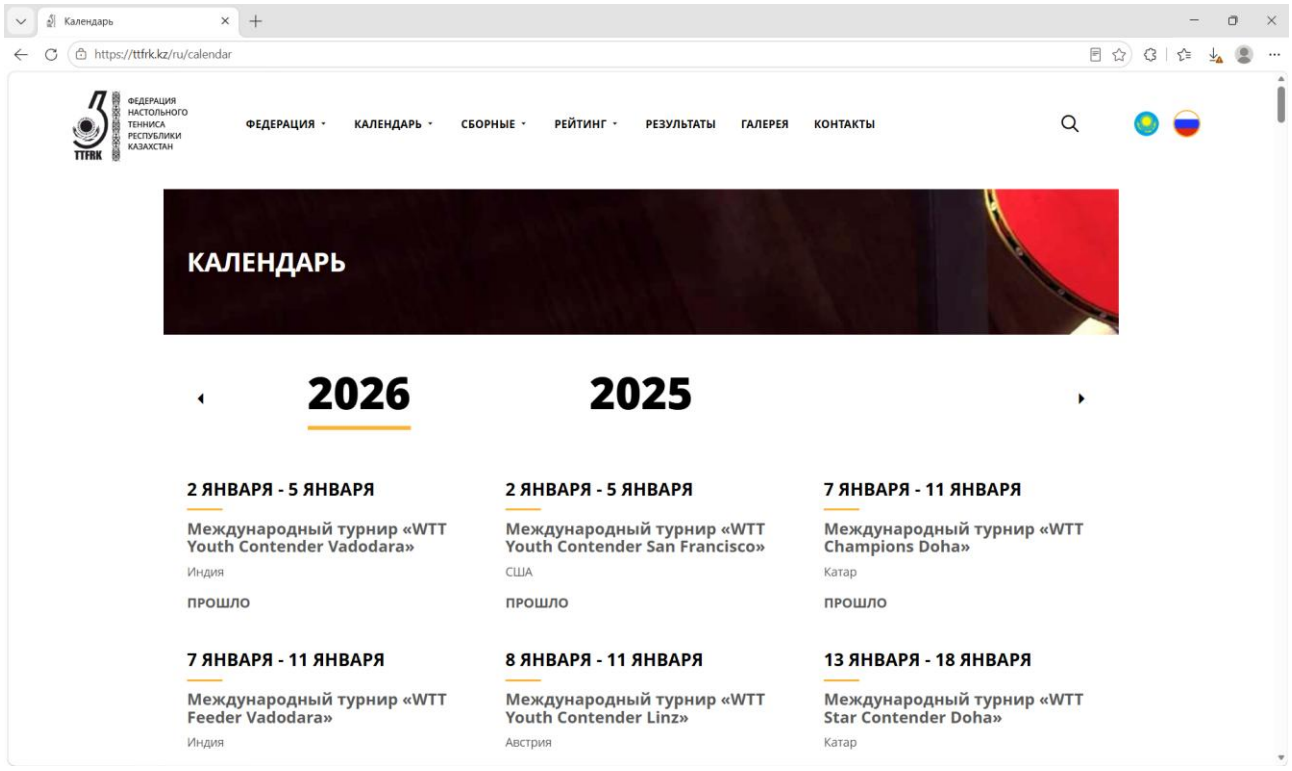


Figure G-01-04. TTFRK official home. Source: https://ttfrk.kz/. Actual capture date: 29.10.2026. Related comparison observations: section 4.

G-05-06. Calendar

https://ttfrk.kz/ru/calendar



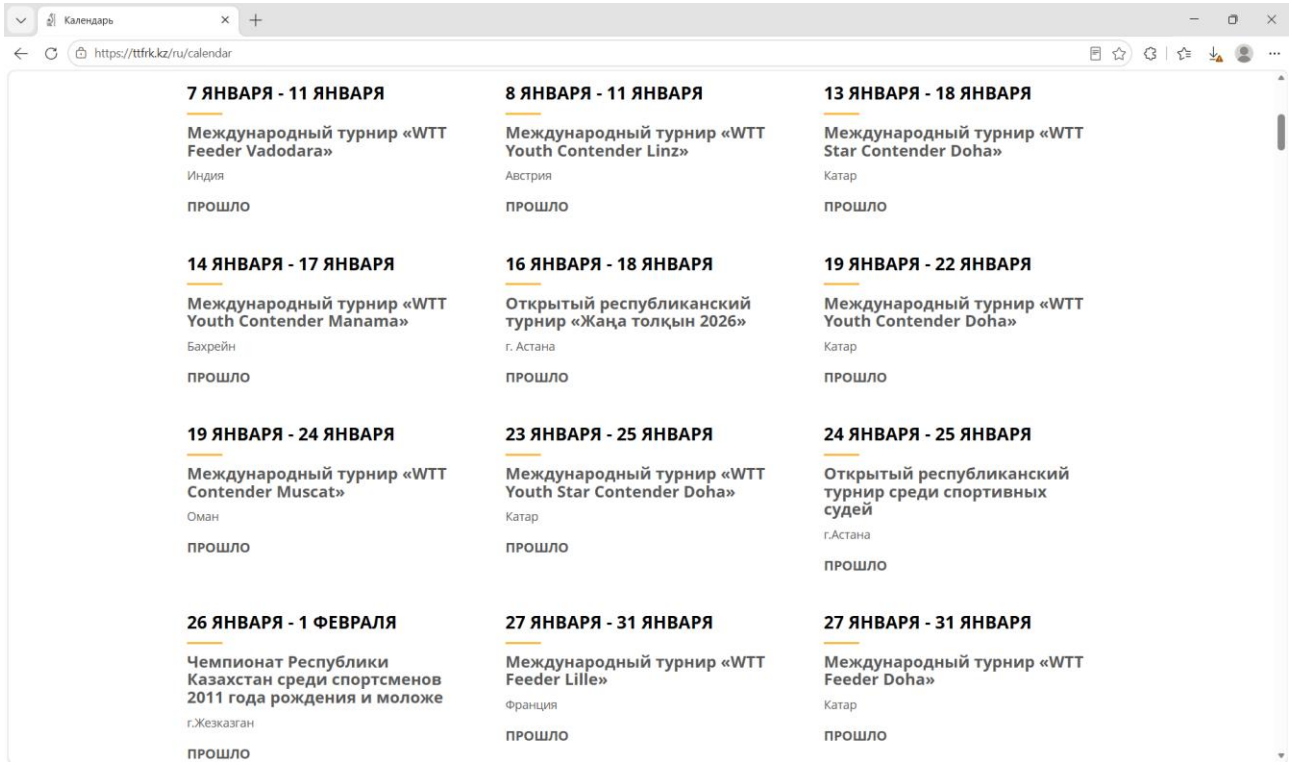
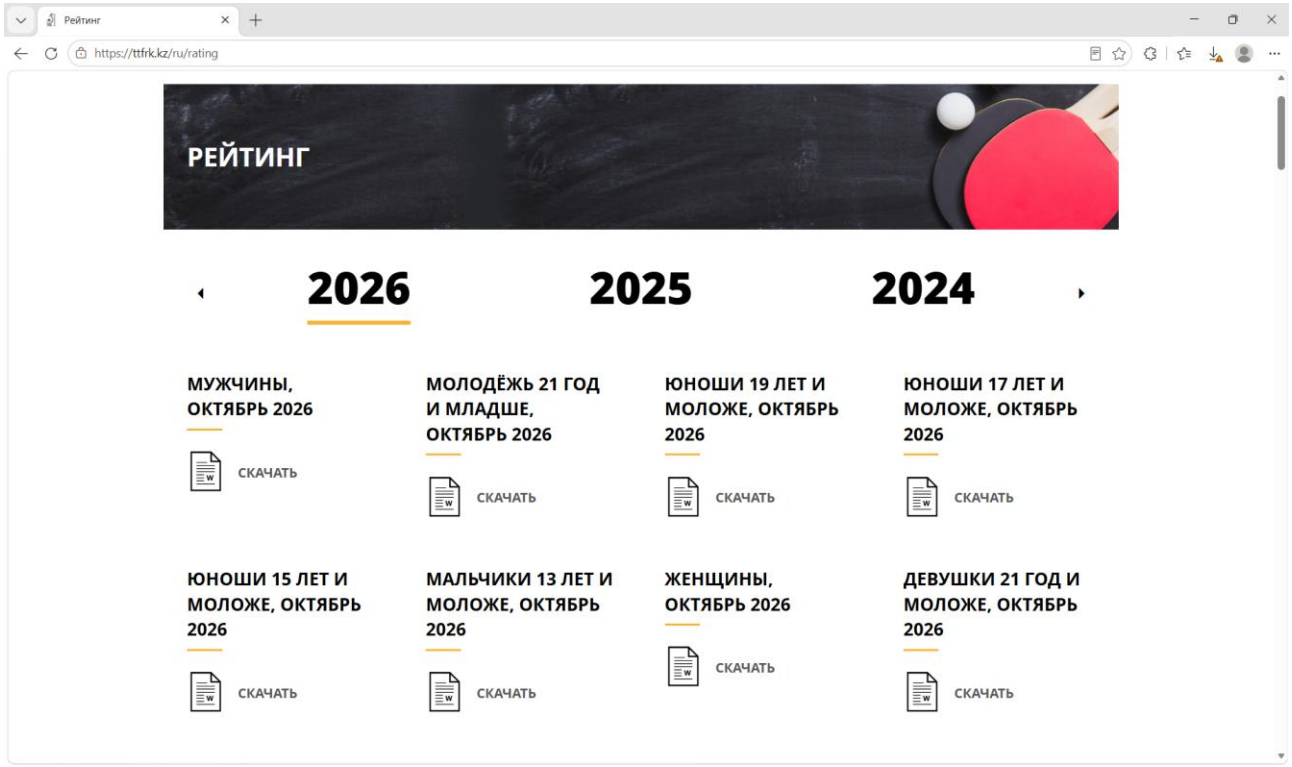
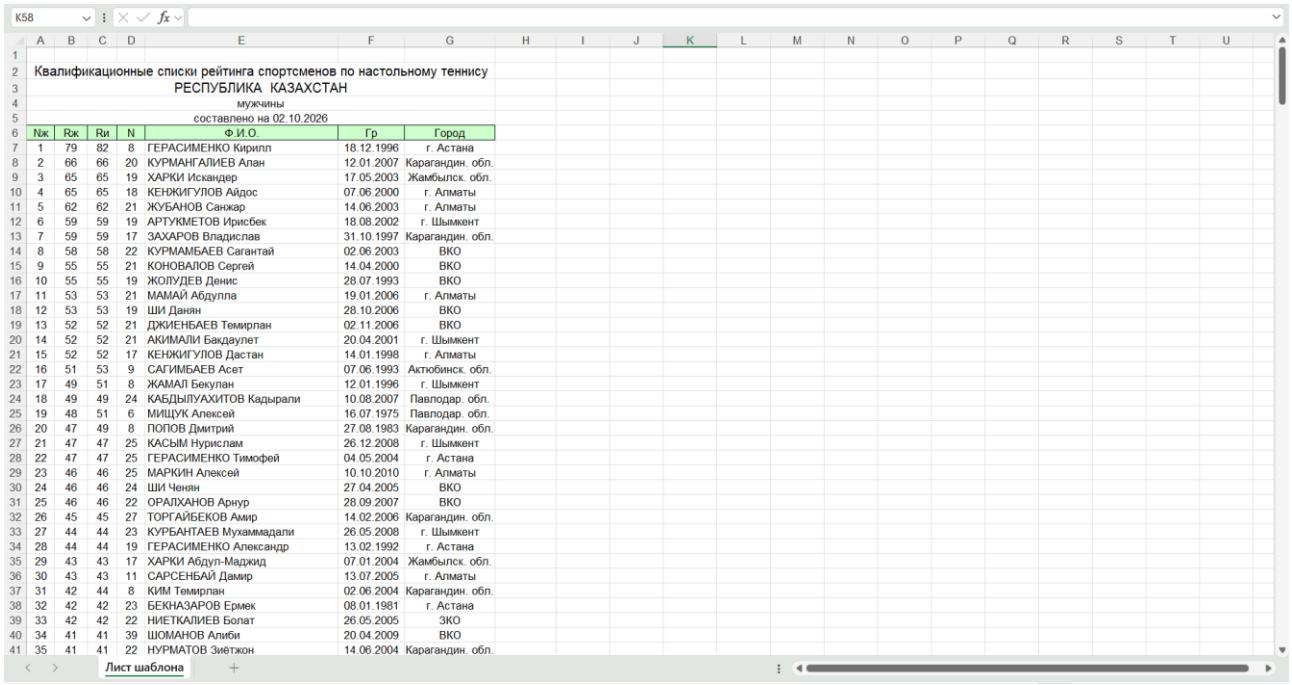


Figure G-05-06. TTFRK calendar. Source: https://ttfrk.kz/ru/calendar. Actual capture date: 29.10.2026. Related comparison observations: section 4.

G-07-08. Rating

https://ttfrk.kz/ru/rating





Квалификационные списки рейтинга спортсменов по настольному теннису
РЕСПУБЛИКА КАЗАХСТАН
мужчины
составлено на 02.10.2026

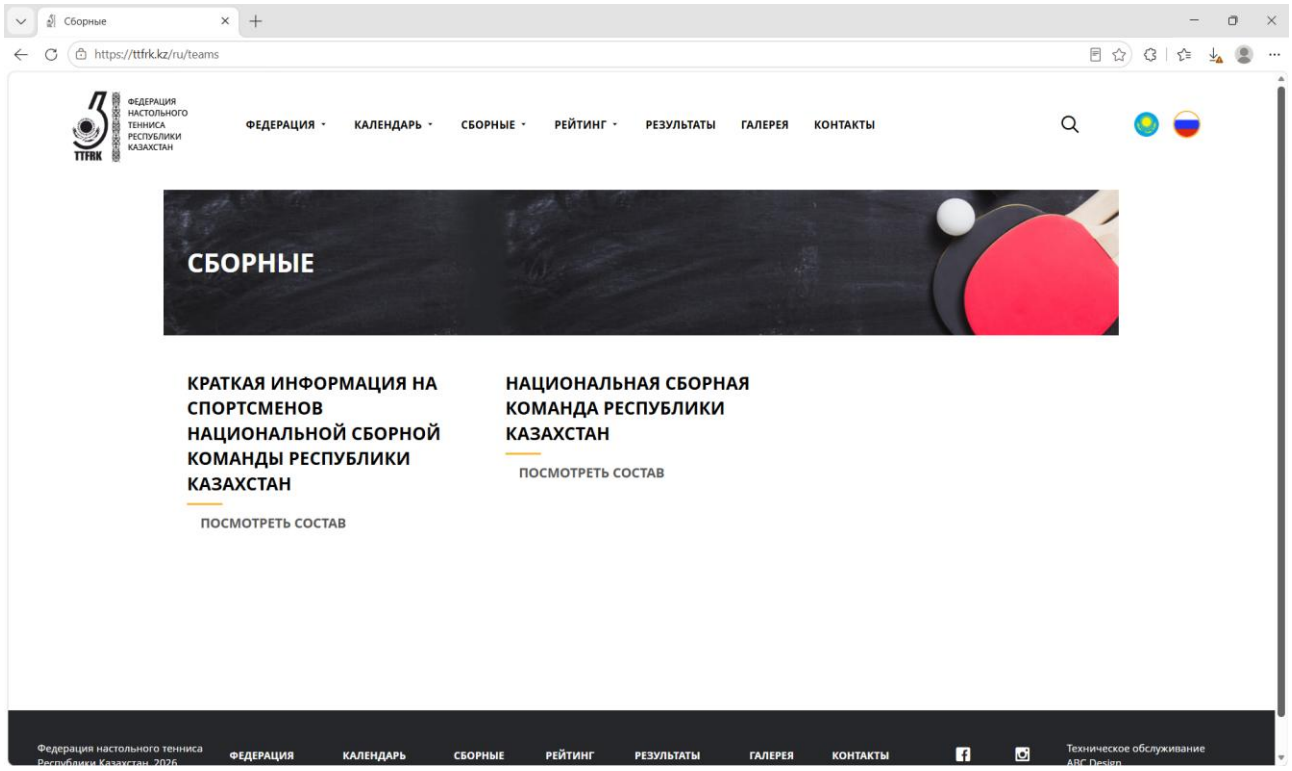
Nж	Rж	Rи	N	Ф.И.О.	Гр	Город
1	79	82	8	ГЕРАСИМЕНКО Кирилл	18.12.1996	г. Астана
2	66	66	20	КУРМАНГАЛИЕВ Алан	12.01.2007	Карагандин. обл.
3	65	65	19	ХАРКИ Искандер	17.05.2003	Жамбылск. обл.
4	65	65	18	КЕНЖИГУЛОВ Айдос	07.06.2000	г. Алматы
5	62	62	21	ЖУБАНОВ Санжар	14.06.2003	г. Алматы
6	59	59	19	АРТУКМЕТОВ Ирисбек	18.08.2002	г. Шымкент
7	59	59	17	ЗАХАРОВ Владислав	31.10.1997	Карагандин. обл.
8	58	58	22	КУРМАМБАЕВ Сагантай	02.06.2003	ВКО
9	55	55	21	КОНОВАЛОВ Сергей	14.04.2000	ВКО
10	55	55	19	ЖОЛУДЕВ Денис	28.07.1993	ВКО
11	53	53	21	МАМАЙ Абдулла	19.01.2006	г. Алматы
12	53	53	19	ШИ Данян	28.10.2006	ВКО
13	52	52	21	ДЖИЕНБАЕВ Темирлан	02.11.2006	ВКО
14	52	52	21	АКИМАЛИ Бакдаулет	20.04.2001	г. Шымкент
15	52	52	17	КЕНЖИГУЛОВ Дастан	14.01.1998	г. Алматы
16	51	53	9	САГИМБАЕВ Асет	07.06.1993	Актюбинск. обл.
17	49	51	8	ЖАМАЛ Бекулан	12.01.1996	г. Шымкент
18	49	49	24	КАБДЫЛУАХИТОВ Кадырали	10.08.2007	Павлодар. обл.
19	48	51	6	МИЩУК Алексей	16.07.1975	Павлодар. обл.
20	47	49	8	ПОПОВ Дмитрий	27.08.1983	Карагандин. обл.
21	47	47	25	КАСЫМ Нурислам	26.12.2008	г. Шымкент
22	47	47	25	ГЕРАСИМЕНКО Тимофей	04.05.2004	г. Астана
23	46	46	25	МАРКИН Алексей	10.10.2010	г. Алматы
24	46	46	24	ШИ Ченян	27.04.2005	ВКО
25	46	46	22	ОРАЛХАНОВ Арнур	28.09.2007	ВКО
26	45	45	27	ТОРГАЙБЕКОВ Амир	14.02.2006	Карагандин. обл.
27	44	44	23	КУРБАНТАЕВ Мухаммадали	26.05.2008	г. Шымкент
28	44	44	19	ГЕРАСИМЕНКО Александр	13.02.1992	г. Астана
29	43	43	17	ХАРКИ Абдул-Маджид	07.01.2004	Жамбылск. обл.
30	43	43	11	САРСЕНБАЙ Дамир	13.07.2005	г. Алматы
31	42	44	8	КИМ Темирлан	02.06.2004	Карагандин. обл.
32	42	42	23	БЕКНАЗАРОВ Ермек	08.01.1981	г. Астана
33	42	42	22	НИЕТКАЛИЕВ Болат	26.05.2005	ЗКО
34	41	41	39	ШОМАНОВ Алиби	20.04.2009	ВКО
35	41	41	22	НУРМАТОВ Зиётжон	14.06.2004	Карагандин. обл.

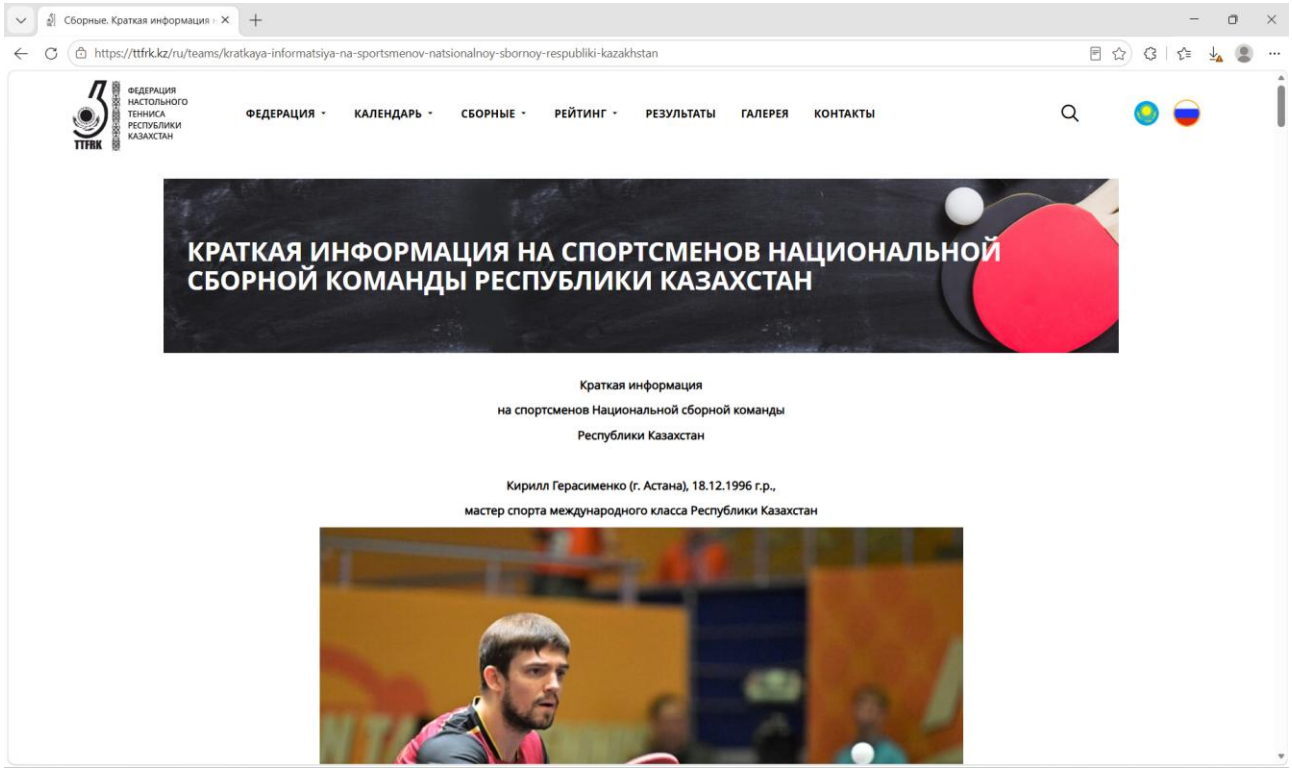
Лист шаблона

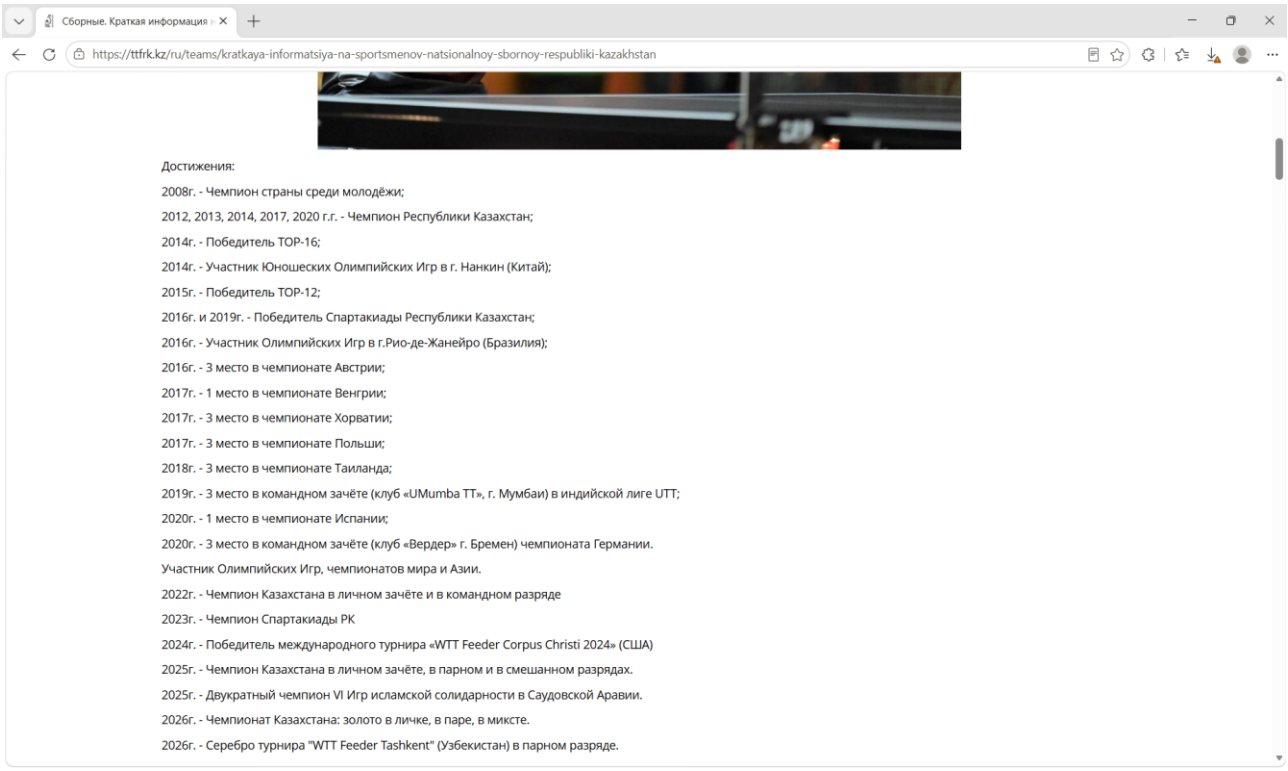
Figure G-07-08. TTFRK rating. Source: https://ttfrk.kz/ru/rating. Actual capture date: 29.10.2026. Related comparison observations: section 4.

G-09-25. National teams

https://ttfrk.kz/ru/teams







Сборные. Краткая информация

https://ttfrk.kz/ru/teams/kratkaya-informatsiya-na-sportsmenov-natsionalnoy-sbornoy-respubliki-kazakhstan

Достижения:

2008г. - Чемпион страны среди молодёжи;

2012, 2013, 2014, 2017, 2020 г.г. - Чемпион Республики Казахстан;

2014г. - Победитель ТОР-16;

2014г. - Участник Юношеских Олимпийских Игр в г. Нанкин (Китай);

2015г. - Победитель ТОР-12;

2016г. и 2019г. - Победитель Спартакиады Республики Казахстан;

2016г. - Участник Олимпийских Игр в г.Рио-де-Жанейро (Бразилия);

2016г. - 3 место в чемпионате Австрии;

2017г. - 1 место в чемпионате Венгрии;

2017г. - 3 место в чемпионате Хорватии;

2017г. - 3 место в чемпионате Польши;

2018г. - 3 место в чемпионате Таиланда;

2019г. - 3 место в командном зачёте (клуб «UMumba TT», г. Мумбаи) в индийской лиге UTT;

2020г. - 1 место в чемпионате Испании;

2020г. - 3 место в командном зачёте (клуб «Вердер» г. Бремен) чемпионата Германии.

Участник Олимпийских Игр, чемпионатов мира и Азии.

2022г. - Чемпион Казахстана в личном зачёте и в командном разряде

2023г. - Чемпион Спартакиады РК

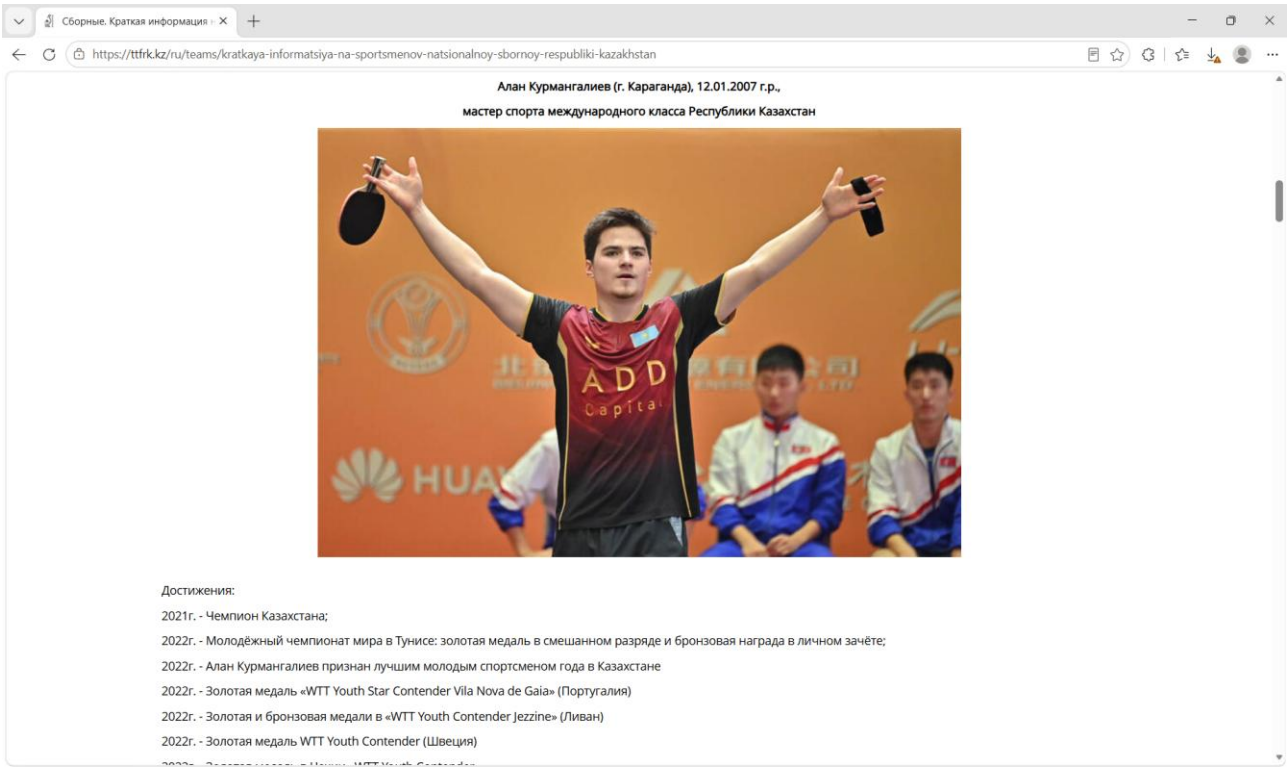
2024г. - Победитель международного турнира «WTT Feeder Corpus Christi 2024» (США)

2025г. - Чемпион Казахстана в личном зачёте, в парном и в смешанном разрядах.

2025г. - Двукратный чемпион VI Игр исламской солидарности в Саудовской Аравии.

2026г. - Чемпионат Казахстана: золото в личке, в паре, в миксте.

2026г. - Серебро турнира "WTT Feeder Tashkent" (Узбекистан) в парном разряде.



Сборные. Краткая информация

https://ttfrk.kz/ru/teams/kratkaya-informatsiya-na-sportsmenov-natsionalnoy-sbornoy-respubliki-kazakhstan

Алан Курмангалиев (г. Караганда), 12.01.2007 г.р.,

мастер спорта международного класса Республики Казахстан

Достижения:

2021г. - Чемпион Казахстана;

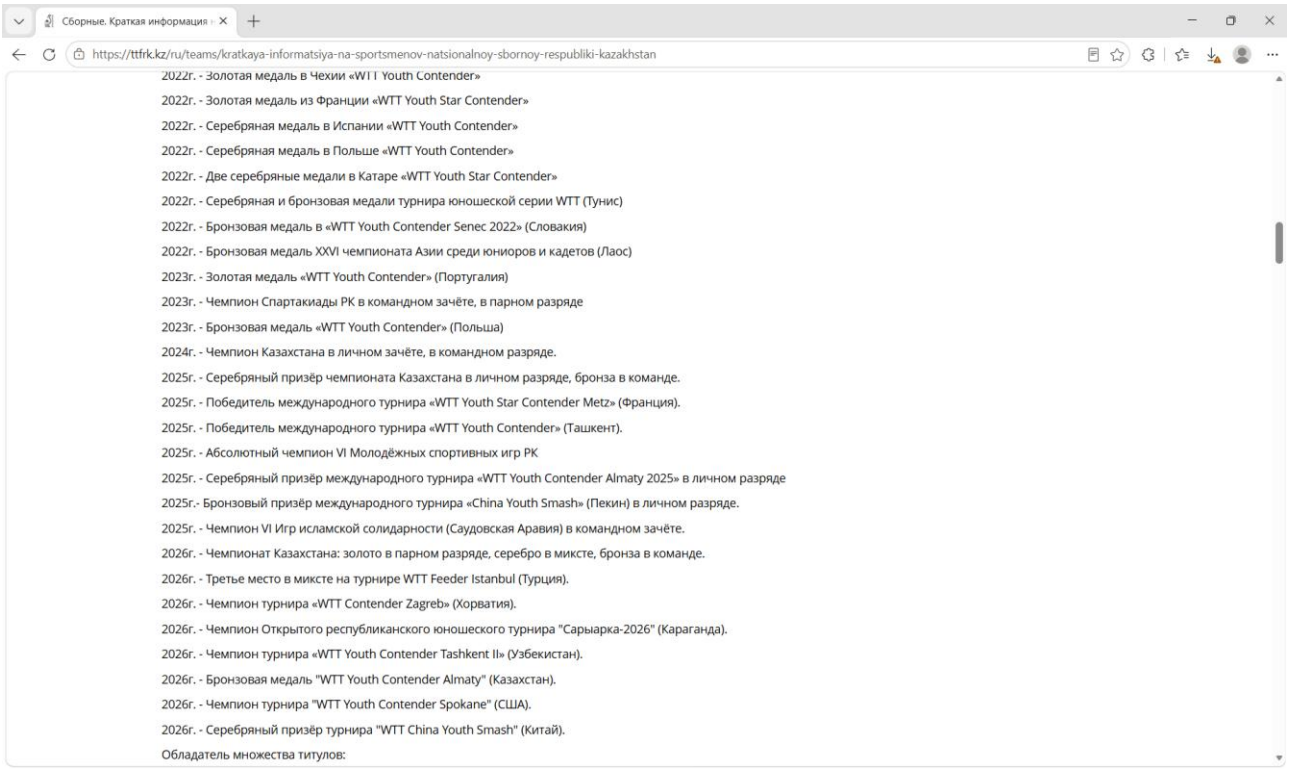
2022г. - Молодёжный чемпионат мира в Тунисе: золотая медаль в смешанном разряде и бронзовая награда в личном зачёте;

2022г. - Алан Курмангалиев признан лучшим молодым спортсменом года в Казахстане

2022г. - Золотая медаль «WTT Youth Star Contender Vila Nova de Gaia» (Португалия)

2022г. - Золотая и бронзовая медали в «WTT Youth Contender Jezzine» (Ливан)

2022г. - Золотая медаль WTT Youth Contender (Швеция)



2022г. - Золотая медаль в Чехии «WTT Youth Contender»

2022г. - Золотая медаль из Франции «WTT Youth Star Contender»

2022г. - Серебряная медаль в Испании «WTT Youth Contender»

2022г. - Серебряная медаль в Польше «WTT Youth Contender»

2022г. - Две серебряные медали в Катаре «WTT Youth Star Contender»

2022г. - Серебряная и бронзовая медали турнира юношеской серии WTT (Тунис)

2022г. - Бронзовая медаль в «WTT Youth Contender Senec 2022» (Словакия)

2022г. - Бронзовая медаль XXVI чемпионата Азии среди юниоров и кадетов (Лаос)

2023г. - Золотая медаль «WTT Youth Contender» (Португалия)

2023г. - Чемпион Спартакиады РК в командном зачёте, в парном разряде

2023г. - Бронзовая медаль «WTT Youth Contender» (Польша)

2024г. - Чемпион Казахстана в личном зачёте, в командном разряде.

2025г. - Серебряный призёр чемпионата Казахстана в личном разряде, бронза в команде.

2025г. - Победитель международного турнира «WTT Youth Star Contender Metz» (Франция).

2025г. - Победитель международного турнира «WTT Youth Contender» (Ташкент).

2025г. - Абсолютный чемпион VI Молодёжных спортивных игр РК

2025г. - Серебряный призёр международного турнира «WTT Youth Contender Almaty 2025» в личном разряде

2025г.- Бронзовый призёр международного турнира «China Youth Smash» (Пекин) в личном разряде.

2025г. - Чемпион VI Игр исламской солидарности (Саудовская Аравия) в командном зачёте.

2026г. - Чемпионат Казахстана: золото в парном разряде, серебро в миксте, бронза в команде.

2026г. - Третье место в миксте на турнире WTT Feeder Istanbul (Турция).

2026г. - Чемпион турнира «WTT Contender Zagreb» (Хорватия).

2026г. - Чемпион Открытого республиканского юношеского турнира "Сарыарка-2026" (Караганда).

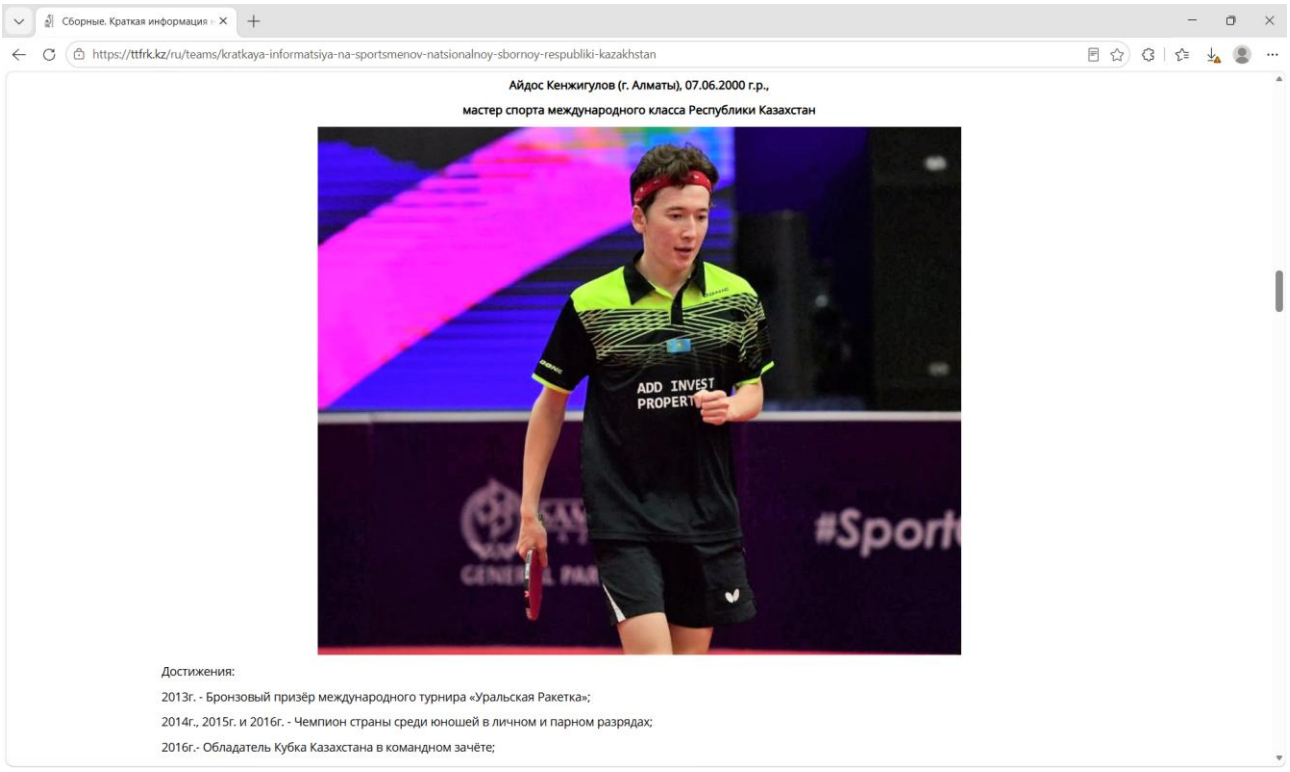
2026г. - Чемпион турнира «WTT Youth Contender Tashkent II» (Узбекистан).

2026г. - Бронзовая медаль "WTT Youth Contender Almaty" (Казахстан).

2026г. - Чемпион турнира "WTT Youth Contender Spokane" (США).

2026г. - Серебряный призёр турнира "WTT China Youth Smash" (Китай).

Обладатель множества титулов:



Айдос Кенжигулов (г. Алматы), 07.06.2000 г.р.,

мастер спорта международного класса Республики Казахстан

Достижения:

2013г. - Бронзовый призёр международного турнира «Уральская Ракетка»;

2014г., 2015г. и 2016г. - Чемпион страны среди юношей в личном и парном разрядах;

2016г.- Обладатель Кубка Казахстана в командном зачёте;

Сборные. Краткая информация

https://ttfrk.kz/ru/teams/kratkaya-informatsiya-na-sportsmenov-natsionalnoy-sbornoy-respubliki-kazakhstan

2017г. - Чемпион страны среди Молодёжи;

2017г. - Победитель турнира ко Дню Президента РК;

2018г. - 3 место в Отборочном турнире на Юношескую Олимпиаду;

2019г. - 2 место в турнире ТОП-12;

2019г. - Чемпион страны среди молодёжи;

2020г. - Победитель турнира ТОП-16;

2020г. - 2 место в турнире ТОП-12;

2020г. – Серебряный призёр чемпионата Казахстана;

2021г. - Чемпион Казахстана в командном зачёте и в смешанной паре,

2024г. - Чемпионат Казахстана: серебряный призёр в личке, в команде, в миксте.

2025г. - Чемпионат Казахстана: серебро в команде, бронза в паре.

2025г. - Обладатель Кубка Австрии в составе команды "Stockerau" (Што́ккерау).

2025г. - Чемпион VI Игр исламской солидарности (Саудовская Аравия) в командном зачёте.

2026г. - Чемпионат Казахстана: золото в команде, серебро в паре, бронза в личке.

2026г. - Кубок Австрии: золото в команде.

Участник чемпионатов мира и Азии.

Санжар Жубанов (г. Алматы), 14.06.2003 г.

мастер спорта Республики Казахстан



Сборные. Краткая информация

https://ttfrk.kz/ru/teams/kratkaya-informatsiya-na-sportsmenov-natsionalnoy-sbornoy-respubliki-kazakhstan



Достижения:

2016 г. - серебряный призёр чемпионата РК среди спортсменов 2003 г.р

2019 г. - 3 место на чемпионате РК среди спортсменов 2001 г.р., чемпион молодежного чемпионата РК среди в парном разряде среди спортсменов 1998 г.р

2021 г. - чемпион РК в командном зачете

2022 г. - Чемпион молодежного чемпионата РК среди спортсменов 2001 г.р

2023 г. - 3 место молодежного чемпионата РК среди спортсменов 2002 г.р

2024 г. - бронзовый призер чемпионата Казахстана в паре, серебряный призер в командном зачёте

2024 г. - бронзовый призер чемпионата РК среди спортсменов 2003 г.р, участник чемпионата мира 2024, участник чемпионата Азии 2024

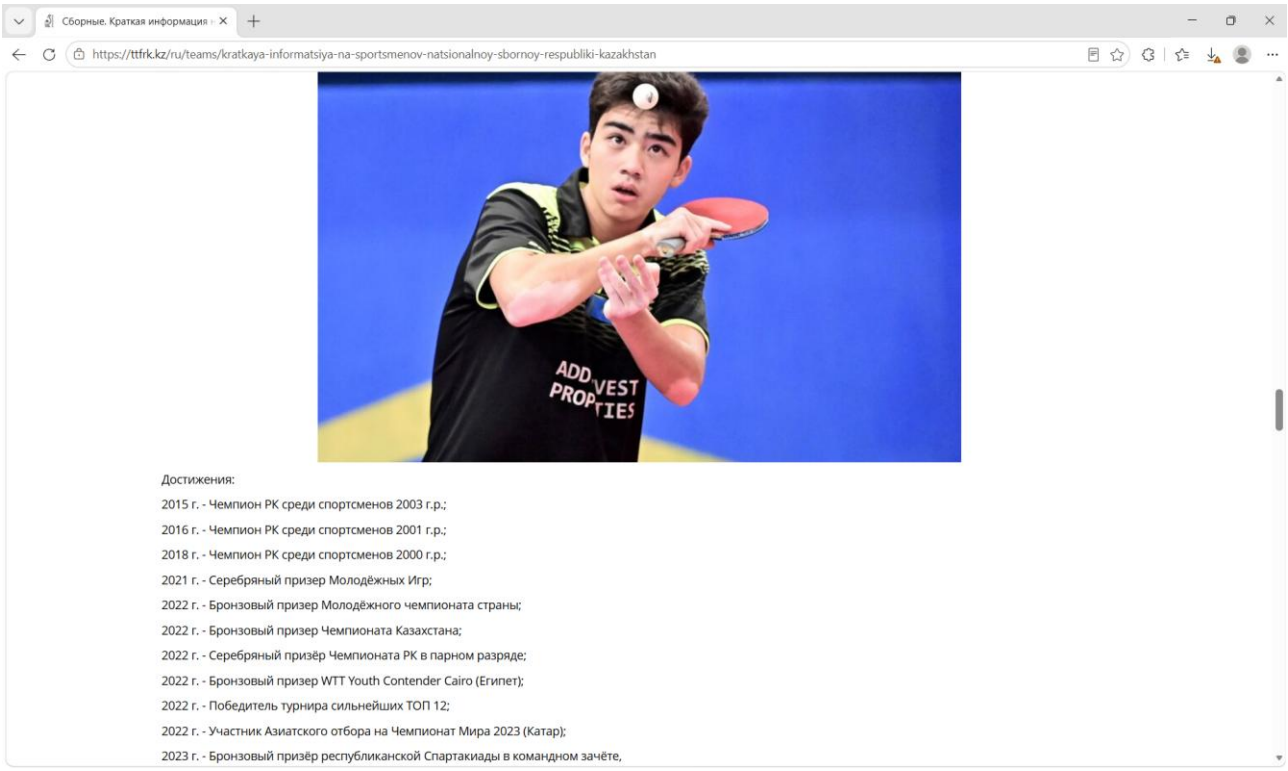
2025 г. - бронзовый призёр чемпионата Казахстана.

2026г. - Чемпионат Казахстана: золото в команде, серебро в личке.

Харки Искендер (Жамбылская область), 17.05.2003 г.

мастер спорта Республики Казахстан





Достижения:

2015 г. - Чемпион РК среди спортсменов 2003 г.р.;

2016 г. - Чемпион РК среди спортсменов 2001 г.р.;

2018 г. - Чемпион РК среди спортсменов 2000 г.р.;

2021 г. - Серебряный призер Молодёжных Игр;

2022 г. - Бронзовый призер Молодёжного чемпионата страны;

2022 г. - Бронзовый призер Чемпионата Казахстана;

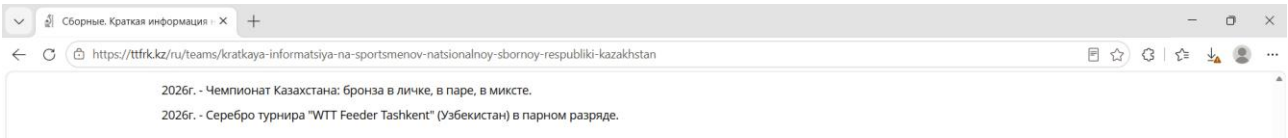
2022 г. - Серебряный призёр Чемпионата РК в парном разряде;

2022 г. - Бронзовый призер WTT Youth Contender Cairo (Египет);

2022 г. - Победитель турнира сильнейших ТОП 12;

2022 г. - Участник Азиатского отбора на Чемпионат Мира 2023 (Катар);

2023 г. - Бронзовый призёр республиканской Спартакиады в командном зачёте,



2026г. - Чемпионат Казахстана: бронза в личке, в паре, в миксте.

2026г. - Серебро турнира "WTT Feeder Tashkent" (Узбекистан) в парном разряде.

Зауреш Акашева (г. Караганда), 02.12.2000 г.р.,

мастер спорта международного класса Республики Казахстан



Сборные. Краткая информация

https://ttfrk.kz/ru/teams/kratkaya-informatsiya-na-sportsmenov-natsionalnoy-sbornoy-respubliki-kazakhstan

Достижения:

2013г. - Абсолютная Чемпионка страны среди девушек 2000 г.р.;

2014г. - Чемпионкой страны среди девушек 1999 г.р.;

2015г. - Абсолютная Чемпионка страны 2000 г.р.;

2016г. - Абсолютная Чемпионка страны 1998 г.р.;

2017г. - 3 место в Чемпионате страны;

2018г. - Победитель турнира ТОП-12

2018г. - 1 место в паре и в команде в чемпионате страны;

2018г. - 2 место в международном турнире «Tunisia Open» в составе интернациональной команды Казахстан-Сингапур;

2018г. - 1 место в международном турнире «Paraguay Open» в составе интернациональной команды Казахстан-Сингапур;

2018г. и 2019г. - Абсолютная чемпионка страны среди молодёжи;

2018г. и 2019г. - Победительница международного турнира «Южная Столица»;

2019г. - 3 место в личном зачёте и 2 в команде в международном турнире «Estonia Open»;

2020г. - Абсолютная чемпионка Казахстана;

2021г. - Чемпионка Казахстана в командном зачёте.

2022г. - Чемпионка Казахстана

2023г. - Чемпионка Спартакиады РК

2024г. - Чемпионка Казахстана в команде, серебряный призёр в личке.

2025г. - Чемпионат Казахстана: золото в личке, в миксте, серебро в команде и в паре.

2025г. - Международный турнир «WTT Feeder» (Спокан, США), третье место в парном разряде с Аной Тофант (Словения).

2025г. - Международный турнир «WTT Feeder Panagyurishte» (Болгария), третье место в личном разряде.

2025г. - Бронзовый призёр турнира «WTT Feeder Gdansk» (Польша).

2026г. - Чемпионат Казахстана: золото в личке, в команде, в миксте.

2026г. - Третье место в миксте на турнире WTT Feeder Istanbul (Турция).

Участница чемпионатов мира и Азии.

Сарвиноз Миркадирова (Алматы), 03.02.2005 г.р.,

Сборные. Краткая информация

https://ttfrk.kz/ru/teams/kratkaya-informatsiya-na-sportsmenov-natsionalnoy-sbornoy-respubliki-kazakhstan

мастер спорта международного класса Республики Казахстан



Достижения:

2017г. - 3 место в международном турнире «Asia Hopes Week»;

2018г. - 3 место в командном зачетё Чемпионата Казахстана;

2018г. - 2 место в международном турнире памяти Акпан Батыра;

2018г. - Победительница международного турнира памяти Роберта Аванесова (г. Андижан);

2018г. - Чемпионат страны среди девушек 2000 г.р.: 1 место в команде, в женской и смешанной паре, 3 место в личном зачёте;

2018г. - Чемпионка страны среди девушек 2003 г.р.;

Сборные. Краткая информация

https://ttfrk.kz/ru/teams/kratkaya-informatsiya-na-sportsmenov-natsionalnoy-sbornoy-respubliki-kazakhstan

2019г. - Чемпионка страны среди девушек 2001 г.р.;

2019г. - Открытое первенство Сербии (ITTF World Junior Circuit): 1 место в личном зачёте и 3 место в команде;

2020г. - Чемпионка Казахстана в парном разряде;

2021г. - Чемпионка Казахстана в личном зачёте.

2022г. - Чемпионка Казахстана в парном разряде

2023г. - Чемпионка Спартакиады РК в парном разряде

2024г. - Чемпионка Казахстана в личном зачёте

2025г. - Чемпионат Казахстана: золото в команде, бронза в личке.

2025г. - Чемпионка VI Молодёжных спортивных игр РК в личном разряде.

2026г. - Чемпионат Казахстана: серебро в команде, бронза в личке, в паре, в миксте.

Анель Бахыт (Алматы), 12.02.2003 г.р.,

мастер спорта международного класса Республики Казахстан



Сборные. Краткая информация

https://ttfrk.kz/ru/teams/kratkaya-informatsiya-na-sportsmenov-natsionalnoy-sbornoy-respubliki-kazakhstan



Достижения:

2018г. - Чемпионка страны среди девушек 2000 г.р.;

2018г. - 3 место в команде и смешанном разряде в чемпионате Казахстана;

2018г. - 2 место в международном турнире памяти Акпан Батыра;

2018г. - 1 место в парном и смешанном разряде, 2 место в команде в чемпионате страны среди девушек 2003г.р.;

2018г. - 3 место в чемпионате страны среди девушек 1997г.р.;

2019г. - 3 место в отборочном турнире в сборную Казахстана;

2019г. - 2 место в республиканском турнире сильнейших 2001 г.р.;

2019г. - 2 место в чемпионате страны среди девушек 1998 г.р.;

2020г. - 2 место в команде и в парном разряде в чемпионате Казахстана;

2021г. - Чемпионка Казахстана в женской паре и в смешанном разряде.

2023г. - Чемпионка Спартакиады РК в парном разряде

2024г. - Серебряный призёр чемпионата страны в команде, бронзовый призёр в личке.

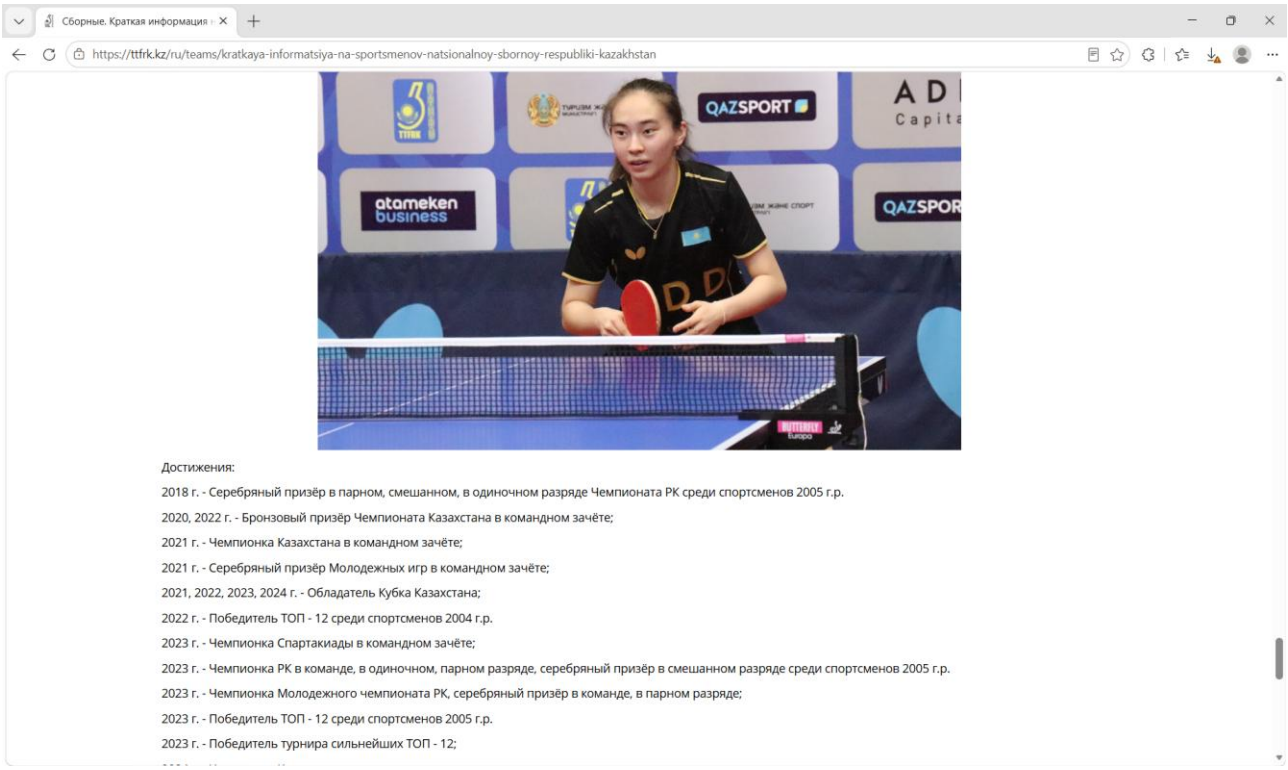
2025г. - Чемпионат Казахстана: золото в команде и в паре.

2026г. - Чемпионат Казахстана: золото в паре, серебро в команде, бронза в личке.

Жанерке Кошкумбаева (г. Караганда), 16.08.2005 г.р.,

мастер спорта Республики Казахстан





Достижения:

2018 г. - Серебряный призёр в парном, смешанном, в одиночном разряде Чемпионата РК среди спортсменов 2005 г.р.

2020, 2022 г. - Бронзовый призёр Чемпионата Казахстана в командном зачёте;

2021 г. - Чемпионка Казахстана в командном зачёте;

2021 г. - Серебряный призёр Молодежных игр в командном зачёте;

2021, 2022, 2023, 2024 г. - Обладатель Кубка Казахстана;

2022 г. - Победитель ТОП - 12 среди спортсменов 2004 г.р.

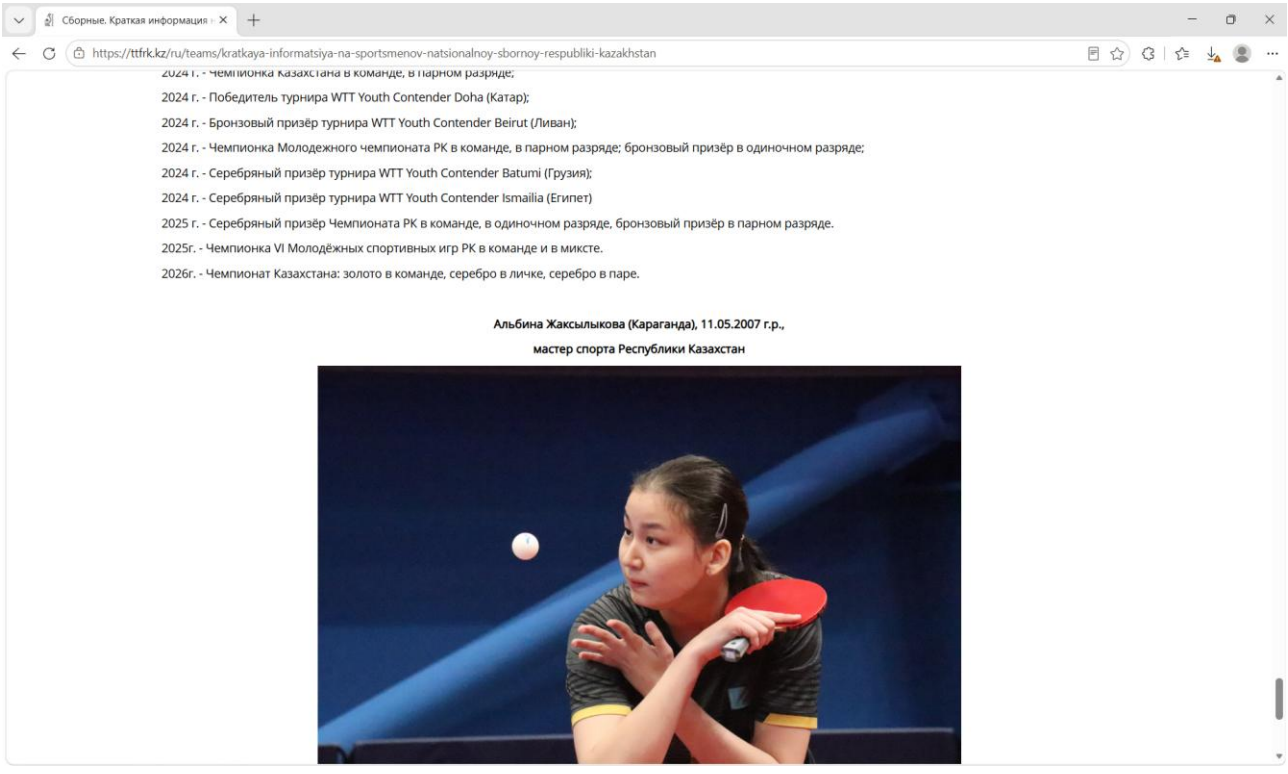
2023 г. - Чемпионка Спартакиады в командном зачёте;

2023 г. - Чемпионка РК в команде, в одиночном, парном разряде, серебряный призёр в смешанном разряде среди спортсменов 2005 г.р.

2023 г. - Чемпионка Молодежного чемпионата РК, серебряный призёр в команде, в парном разряде;

2023 г. - Победитель ТОП - 12 среди спортсменов 2005 г.р.

2023 г. - Победитель турнира сильнейших ТОП - 12;



2024 г. - Чемпионка Казахстана в команде, в парном разряде;

2024 г. - Победитель турнира WTT Youth Contender Doha (Катар);

2024 г. - Бронзовый призёр турнира WTT Youth Contender Beirut (Ливан);

2024 г. - Чемпионка Молодежного чемпионата РК в команде, в парном разряде; бронзовый призёр в одиночном разряде;

2024 г. - Серебряный призёр турнира WTT Youth Contender Batumi (Грузия);

2024 г. - Серебряный призёр турнира WTT Youth Contender Ismailia (Египет)

2025 г. - Серебряный призёр Чемпионата РК в команде, в одиночном разряде, бронзовый призёр в парном разряде.

2025г. - Чемпионка VI Молодёжных спортивных игр РК в команде и в миксте.

2026г. - Чемпионат Казахстана: золото в команде, серебро в личке, серебро в паре.

Альбина Жаксылыкова (Караганда), 11.05.2007 г.р.,

мастер спорта Республики Казахстан

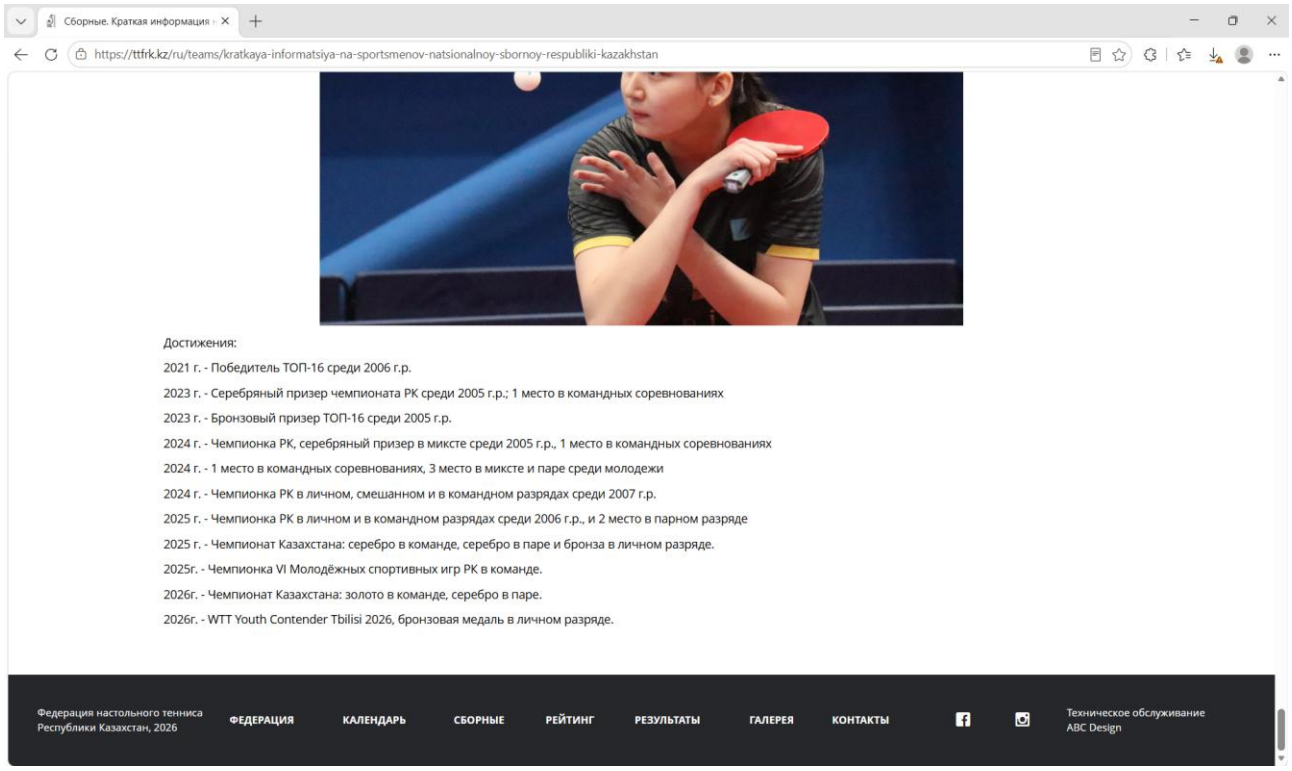


Figure G-09-25. TTFRK national teams. Source: https://ttfrk.kz/ru/teams. Actual capture date: 29.10.2026. Related comparison observations: section 4.

Appendix H Editable diagram sources and exports

The editable sources and PNG exports correspond to the diagrams in Appendix C.

C.1 System Context Diagram
Editable source: docs/diagrams/source/system-context.drawio
PNG export: docs/diagrams/exports/system-context-1.png

C.2 Use Case Diagram
Editable source: docs/diagrams/source/use-case.drawio
PNG export: docs/diagrams/exports/use-case-2.png

C.3 Component Diagram
Editable source: docs/diagrams/source/component.drawio
PNG export: docs/diagrams/exports/component-3.png

These diagrams describe the proposed architecture and do not demonstrate implemented backend functionality.